Chapter 5
Flow Measurement

5.1 Principles of Flow Measurement

5.1.1 Types of Flow

There are four main types of liquid flow:

Type of flow	Notable Characteristics	Process material
General	Thin, clean liquids	Water, light oils and solvents
Two-phase flow	Liquids with bubbles	Beer, wet steam, unrefined petroleum
Slurry	Dirty liquids	Water and sand
Non-Newtonian	Heavy thick liquids	Grease, Paint, honey

Table 5.1
Main types of liquid flow

5.1.2 Basic Terms and Concepts

Laminar vs Turbulent flow

In laminar flow, the fluid moves smoothly in orderly layers, with little or no mixing of the fluid across the flow stream. With laminar flow, there can still exist changes in velocity as the friction of the wall slows the layers closest to the wall, while the flow in the centre of the pipe moves at a faster pace. This velocity change produces a parabolic streamlined flow profile.

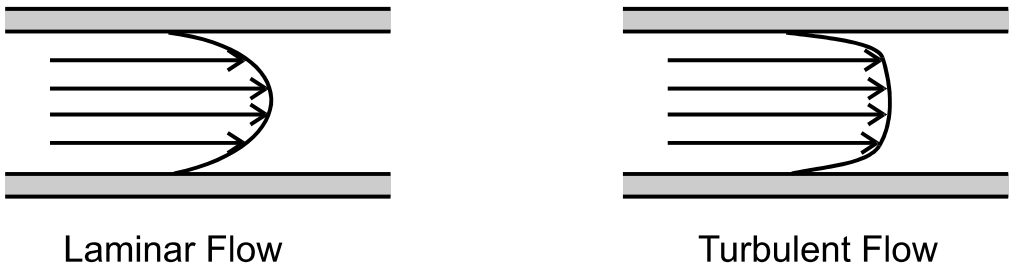


Figure 5.1
Flow profile

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In turbulent flows, the laminar flow breaks down to produce intermixing between the layers. Turbulent flow is quite random, as smaller currents flow in all directions - these are also known as eddies. This type of flow has a flatter flow profile, such that the velocity of forward flow in the centre of the pipe is nearly the same as that near the walls of the pipe.

Swirl

Swirl occurs with laminar flows as fluid passes through elbows or some other form of pipeline geometry. In a similar fashion to turbulent flows, they affect the measurements of many instruments and precautions should be taken to mount measuring devices well downsteam from the swirling fluid. The effects of swirls can be minimised by the use of a flow conditioner, or straightener in the line upstream.

Reynolds number

A Reynolds number defines the flow conditions at a particular point. It is a way of representing fluidity and is a useful indicator of laminar and turbulent flow. Laminar flow exists if the Reynolds number is less than 2000, and turbulence when the number is above 4000. There is not a clear transition between laminar and turbulent flows, which does complicate flow measurement in this range of operation.

The Reynolds number equation shown below shows the relationship between the density (ρ), viscosity (u_{cp}), pipe inside diameter (D) and the flow rate (v).

$$RD = \frac{K \times \rho \times v \times D}{u_{cp}}$$

Velocity

This is the speed at which the fluid passes a point along the pipe. The velocity is used to calculate volume and mass flow rates.

Volumetric flow rate

The volumetric flow rate represents that volume of fluid which passes through a pipe per unit of time. This form of measurement is most frequently achieved by measuring the velocity of a fluid with a DP sensor as it travels through a pipe of known cross sectional area.

Mass flow rates

Mass flow is a measure of the actual amount of mass of the fluid that passes some point per unit of time

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Totalized flow

Totalized flow is an accumulation of the amount of fluid that has passed a point in a pipe. This is used in both volume or mass based systems generally for accounting purposes.

Viscosity

The viscosity of a fluid defines the resistance hich the fluid has against flowing. It is typically measured in units of poise or centipoise at a given temperature. Higher numbers indicate an increase in viscosity (which is a greater resistance to flow). As the temperature of most liquids increases, their viscosity decreases and they flow more readily.

Density

The density of a fluid is the mass of the fluid per unit of volume. Density is affected by temperature and pressure variations. To avoid problems of pressure and temperature variation, mass flow measurements can be used. Although density can be of some use, it is usually more often used to infer composition, concentration of chemicals in solution or solids in suspension.

Specific Gravity

The specific gravity is the unitless ratio, found by comparing the density of a substance to the density of water at a standard temperature.

Pulsation's

In maintaining good measuring conditions, pulsations in the flow stream should be recognised and avoided if possible. Common causes of pulsating flows are from pumps, compressors and two-phase fluids.

Cavitation

This condition occurs when the pressure is reduced below the vapour pressure of the liquid. This causes vapour cavities, or bubbles, to form. These bubbles can dissipate quickly causing damage to the piping system as well as the flowmeter, resulting in measurement errors. Maintaining sufficient pressure in the pipe commonly solves the problem. In applications where this is likely, measurement methods requiring a pressure drop should be avoided, such as differential pressure devices.

Non-Newtonian

The normal behaviour of fluids is such that as temperature increases, viscosity decreases. There are fluids that do not abide by the standard rules of fluid dynamics. The so-called non-Newtonian fluids have viscosities that change with shear rate. Some products become thinner when agitated, whereas others stiffen when deformed.

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Vena Contracta

The cross-sectional area of a fluid decreases as it is forced through a restriction. With the inertia of the fluid, the cross-sectional area continues to decrease after the fluid has passed the restriction. Because of this, the fluids minimum cross-sectional area, which also has the maximum velocity and lowest pressure, is located at some point downstream of the restriction. This point is called the vena-contracta.

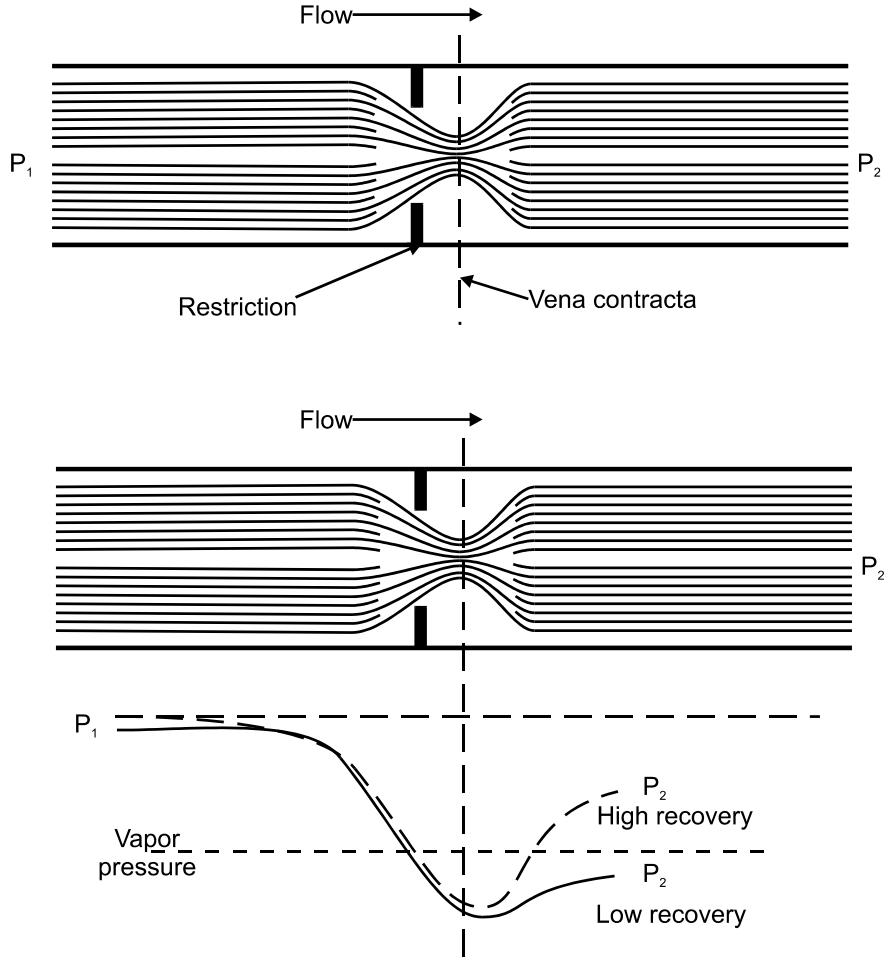


Figure 5.2
Vena-Contracta
(Courtesy of Fisher Controls International)

Rangeability

Rangeability is the ratio of maximum to minimum flow. Rangeability is used to give a relative measure of the range of flows that can be controlled.

$$Rangeability = \frac{Q\max}{Q\min}$$

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5.1.3 Flow Measurements

There are three main measurements that can be made with process flows:

- Velocity
- Volumetric flow
- Mass flow

Velocity is the speed at which the fluid moves. This by itself does not give any information about the quantity of fluid.

Volumetric flow is often deduced by knowing the cross sectional area of the fluid. Most volumetric flow equipment measures the velocity and calculates the volumetric flow based on a constant cross sectional area.

$Q = v \cdot A$

where: v is the velocity
A is the cross sectional area
Q is the volumetric flow rate

Mass flow rate can only be calculated from the velocity or the volumetric flow rates if the density is constant. If the density is not constant, then mass flow measuring equipment is required for mass flow rate.

$W = Q \cdot \rho$

where: Q is the volumetric flow rate
ρ is the density of the fluid
W is the mass flow rate

The flow of gases is normally measured in terms of mass per unit time. While most liquids are nearly incompressible, densities of gases vary with operating temperature and pressure. Some flowmeters, such as Coriolis meters, measure the mass flow directly. Volumetric flowmeters do not measure mass flow directly. Mass flow is calculated from the density and the volumetric flow as shown above. Some volumetric meters infer density based on the measured pressure and temperature of the fluid. This type of measurement is referred to as the inferred method of measuring mass flow.

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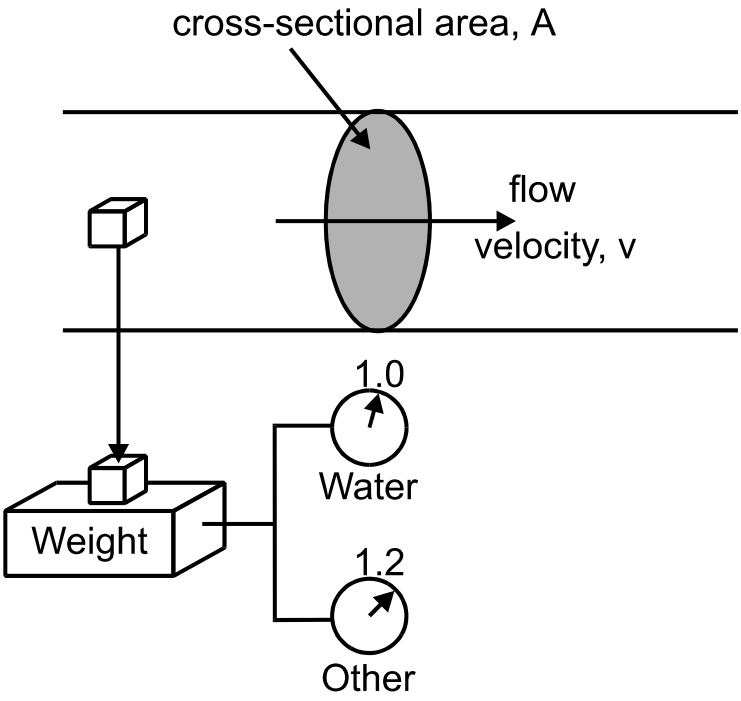


Figure 5.3
Inferred method of measuring mass flow

5.2 Differential Pressure Flowmeters

5.2.1 Basis of Operation

One of the most common methods of measuring flow is with a Differential Flowmeter. This technique requires the pressure to be measured on both sides of an imposed restriction in the path of normal flow. The flow rate of the material can be calculated from the change in pressure.

Before looking at the mathematics, some terms that best describe the response of imposing a restriction in a line need to be considered.

As an example, consider a garden hose with a spray nozzle. With the spray shut off (so that no water flows) the pressure in the hose can be felt. At this point, the hose has a lot of potential energy (or stored energy) - it has the ability to move the water if allowed to do so. When the spray nozzle is opened, energy is converted into a flow of water, ultimately converting the potential energy to a small amount of kinetic energy (or motion energy). If the nozzle is opened fully then most of the potential energy is converted to kinetic energy and the pressure on the hose drops considerably.

Differential pressure devices work on the principle of inducing a change in pressure by placing a restriction in the line. This effectively changes some potential energy kinetic energy - this is detected and measured as a change in pressure.

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The principle of increasing kinetic energy by sacrificing potential energy when constricting the flow is described by mathematics (according to Bernoulli's streamline energy equation).

The restriction in the pipe in called the primary element. The secondary element is the differential pressure sensor and transmitter. This device can measure the and calculate the flow rate. Differential pressure or flow rate information can then be accessed from this device.

The velocity of flow is related to the square root of the differential pressure.

When the pressure differential is measured, the volumetric or mass flow rate can be calculated based on:

- fluid properties
- cross-sectional area
- shape of restriction
- adjacent piping

5.2.2 Formulae

The relationship between the flow rate and the change in pressure can be shown as:

$$V = k.\sqrt{h/\rho}$$

$$Q = k.A.\sqrt{h/\rho}$$

$$W = k.A.\sqrt{h\rho}$$

where: v is the velocity
Q is the volumetric flow rate
W is the mass flow rate
k is a constant
h is the differential pressure
ρ is the density of the fluid
A is the cross sectional area of the pipe

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5.2.3 Primary Element

Orifice Plate

A standard orifice plate is simply a smooth disc with a round, sharp-edged inflow aperture and mounting rings. In the case of viscous liquids, the upstream edge of the bore can be rounded. The shape of the opening and its location do vary widely, and this is dependent on the material being measured. Most common are concentric orifice plates with a round opening in the centre. They produce the best results in turbulent flows when used with clean liquids and gases.

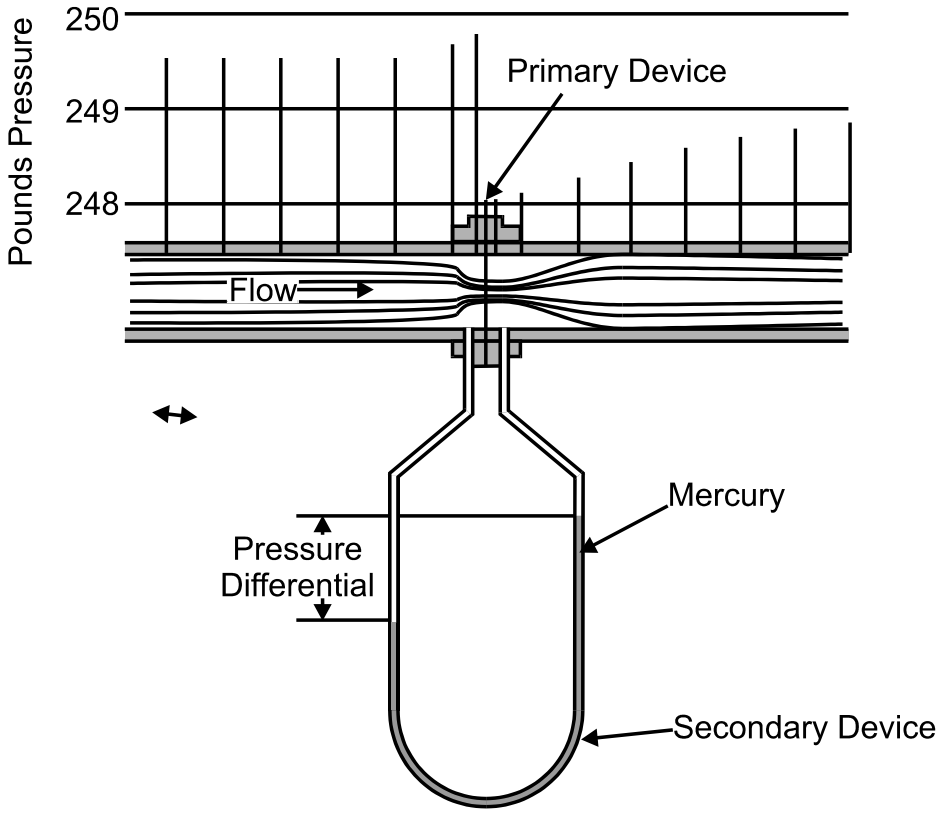


Figure 5.4
Differential pressure profile with orifice plate

Options are available for unclean material or flows that contain liquid and gas of varying densities. When measuring liquids the bore can be positioned at the top of the pipeline to allow the passage of gases. The same applies when allowing suspended solids to pass, by positioning the bore at the bottom and gaining a more accurate liquid flow measurement. Half-circle bores are often used with light slurries or dirty gases.

The pressure removal holes or slits are located in the aperture and mounting rings. The location of these pressure taps depends on the type of tap being used. However, the taps are usually located in the adjacent flanges, or one diameter upstream and one-half diameter downstream.

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Standard orifice meters are primarily used to measure gas and vapour flow. Measurement is relatively accurate. Because of the obstruction of flow there is a relatively high, residual permanent pressure loss. These meters are well understood, rugged and relatively inexpensive for large pipe sizes and are suited for most clean fluids and are not influenced by high temperatures.

Orifice Type

There are two main types of orifices for various applications:

- Concentric, square edged
- Concentric, quadrant edged
- Eccentric or segmental square edged

Concentric, square edged

This is the most common and basic type of orifice meter. This device is typically a thin concentric sharp-edged orifice plate. Because of the simplicity, it is inexpensive to manufacture to very close tolerances. This also simplifies the ease in which it can be installed and replaced.

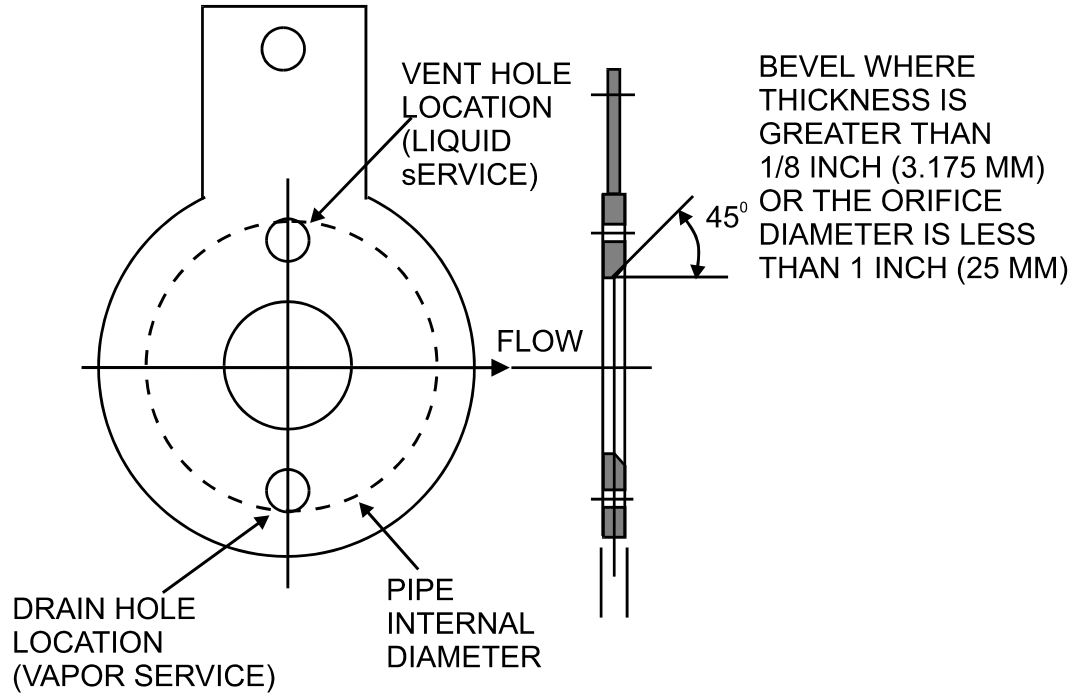


Figure 5.5
Concentric orifice plate

The thickness of the orifice plate depends on the following:

- Pipe size
- Flow temperature
- Differential pressure

Concentric, quadrant edged

This type of orifice plate is used to give increased stability in flow, and is about 10 times that for conventional plates.

Flange taps are typically used with this type of primary element.

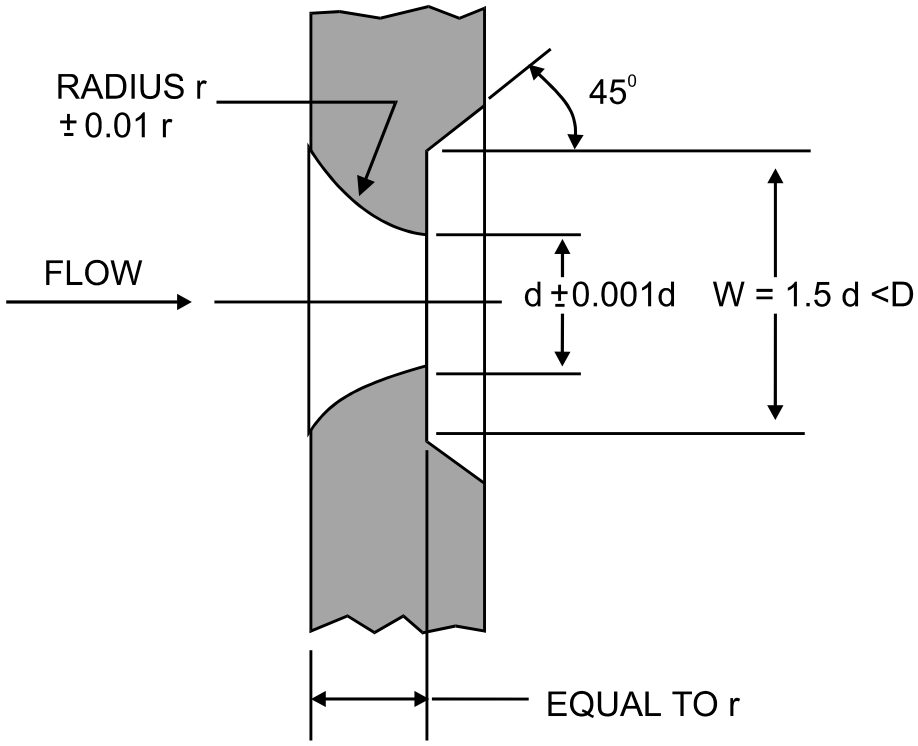


Figure 5.6
Quadrant edge orifice plate

Eccentric or segmental square edged

These are generally used when the process material contains foreign matter that may block the orifice in the case of a concentric configuration.

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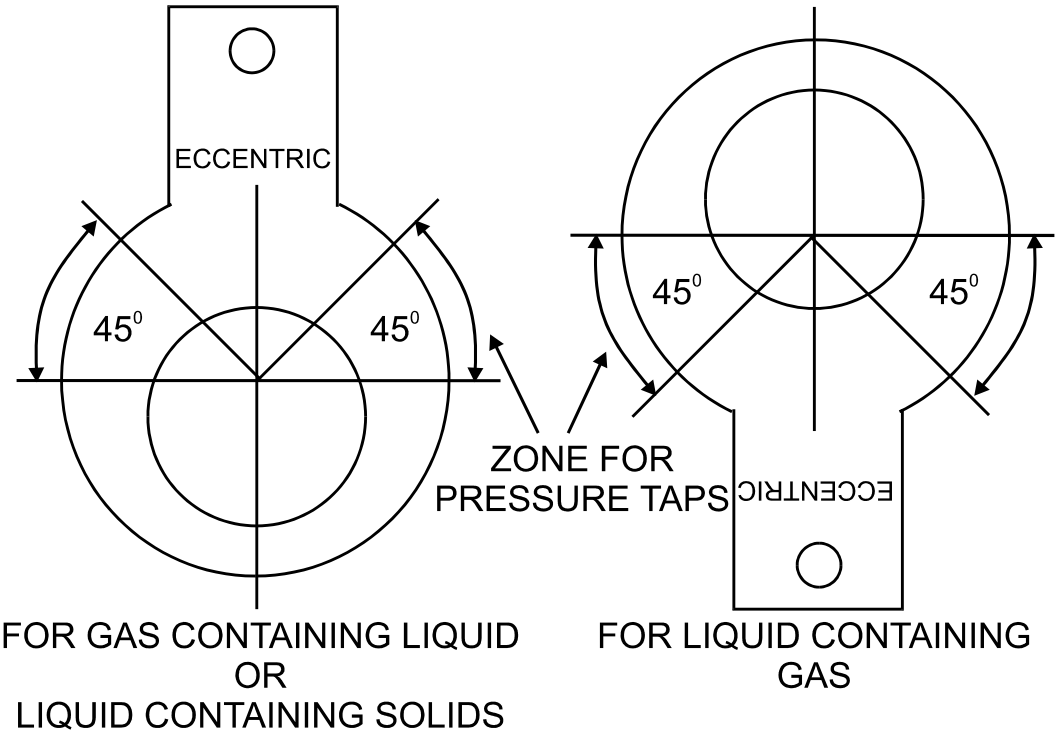


Figure 5.7
Segmental orifice plate

5.2.4 Tap Locations

There are five common locations for differential pressure taps:

- Flange (>50mm)
- Vena contracta (>150mm)
- Radius
- Pipe (full flow)
- Corner taps (<50mm)

Flange Taps

Flange taps are quite common and are generally used for pipe sizes of 50mm and greater. Flange taps are a convenient alternative to drilling and tapping the main pipe for pressure connections.

Note that flow for several pipe diameters beyond the vena contracta is turbulent and does not provide suitable pressure measurements. It is for this reason that flange taps are not used for pipe diameters less than 50mm, as the vena contracta starts to become close to and possible forward of the downstream tapping point.

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Vena contracta taps

Vena contracta taps are limited to pipe sizes greater than 150mm. This limitation is mainly imposed to provide adequate clearance of the tap from the flange.

The vena contracta is where the fluid flow has the smallest cross-sectional area, and also has the lowest pressure. Vena contracta taps are designed around achieving the maximum differential pressure.

The vena contracta is dependent on the flow rate and pipe size, and can vary should either of these parameters change. The vena contracta taps will therefore be affected should the flow rate or pipe size change.

Pipe Taps

Pipe taps are located 2.5 pipe diameters upstream and 8 pipe diameters downstream of the orifice plate.

Pipe taps are used typically in existing installations, where radius and vena contracta taps cannot be used. They are also used in applications of greatly varying flow as the measurement is not affected by flowrate or orifice size.

Accuracy is reduced as they do not measure the maximum available pressure.

Corner taps

Corner taps measure the pressure in the corner between the orifice plate and the pipe wall. Uses for corner taps are found in installations with pipe diameters less than 50mm.

Radius taps

Radius taps are a modification on the vena contracta taps, where the downstream tap is located one-half pipe diameter from the orifice plate. This is to avoid the unstable region that occurs immediately after the orifice plate. Radius taps are generally preferred to vena contracta as the pressure tap location is simplified.

5.2.5 Selection and Sizing

General sizing procedure:

1. First determine the equivalent flow rate corresponding to the process flow rate:

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To convert liquids to equivalent water flow, use

$$q_e, water = (q_{lpm})(G_b)\sqrt{1/G_f}$$

where:

$q_{e,\ water}$	is the Equivalent liquid flow rate
q_{lpm}	Liquid flow rate in litres per minute
G_b	Liquid specific gravity at Base conditions
G_f	Liquid specific gravity at flowing conditions

2. Select the applicable graph based on fluid type, flow units and orifice bore size.

3. Find the calculated flow value on the horizontal axis of the graph. Follow the vertical line until it intersects with the desired orifice bore line. The horizontal line intersecting that point indicates the corresponding differential pressure.

4. Once the correct graph and line are chosen, any combination of equivalent flow rate and differential pressure can be found on the graph.

5.2.6 Advantages

- Simple construction.
- Inexpensive.
- Easily fitted between flanges.
- No moving parts.
- Large range of sizes and opening ratios.
- Suitable for most gases and liquids.
- Well understood and proven.
- Price does not increase dramatically with size.

5.2.7 Disadvantages

- Inaccuracy, typically 1%.
- Low Rangeability, typically 4:1.
- Accuracy is affected by density, pressure and viscosity fluctuations.
- Erosion and physical damage to the restriction affects measurement accuracy.
- Cause some unrecoverable pressure loss.
- Viscosity limits measuring range.
- Require straight pipe runs to ensure accuracy is maintained.
- Pipeline must be full (typically for liquids).

5.2.8 Application Limitations

The inaccuracy with orifice type measurement is due mainly to process conditions and temperature and pressure variations. Ambient conditions and upstream and

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downstream piping also affect the accuracy because of changes to the pressure and continuity of flow.

Standard concentric orifice plate devices should not be used for slurries and dirty fluids, or in applications where there is a high probability of solids accumulating near the plate. Half-circle or eccentric bores can be used for these applications.

5.2.9 Venturi Tube

In a venturi tube, the fluid is accelerated through a nozzle shaped inflow piece (converging cone) which induces a local pressure drop. After passing through the cylindrical restriction, it is released through an expanding section (diffuser) where it returns the flow to near its original pressure.

The Venturi Tube is often selected because pressure drop is not as significant as with the orifice plate and accuracy is better maintained.

Due to the relatively high cost of the Venturi Tube, applications are generally limited to high flow rate fluids, such as main steam lines.

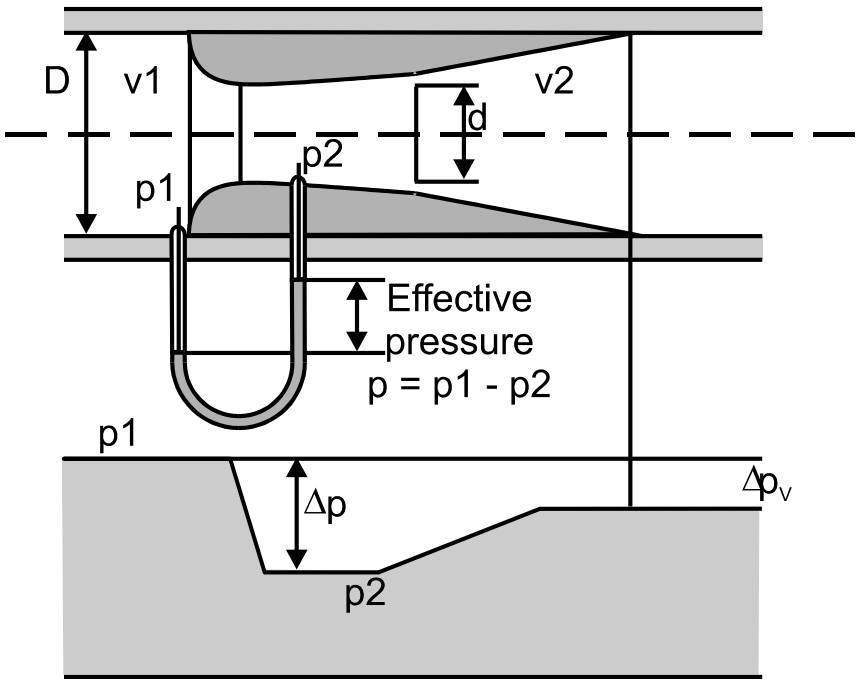


Figure 5.8
Flow measurement with a restriction

The advantages and disadvantages highlight any differences from the orifice meter and are over and above those listed for that device.

Advantages

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- Less significant pressure drop across restriction.
- Less unrecoverable pressure loss.
- Requires less straight pipe up and downstream.

Disadvantages

- More expensive.
- Bulky - requires large section for installation.

5.2.10 Flow Nozzles

The Flow Nozzle is similar to the venturi but are in the shape of an ellipse. They have a higher flow capacity than orifice plates. Another main difference between the flow nozzle and the venturi is that although they have similar inlet nozzles, the flow nozzle has no exit section.

Flow Nozzles can handle larger solids and be used for higher velocities, greater turbulence and high temperature applications. They are often used with fluid or steam applications containing some suspended solids, and in applications where the product is being discharged from service.

These devices are more cost effective, but as such they provide less accuracy than venturis, and have a higher unrecoverable pressure loss.

The advantages and disadvantages highlight any differences from the venturi tube and are over and above those listed for that device.

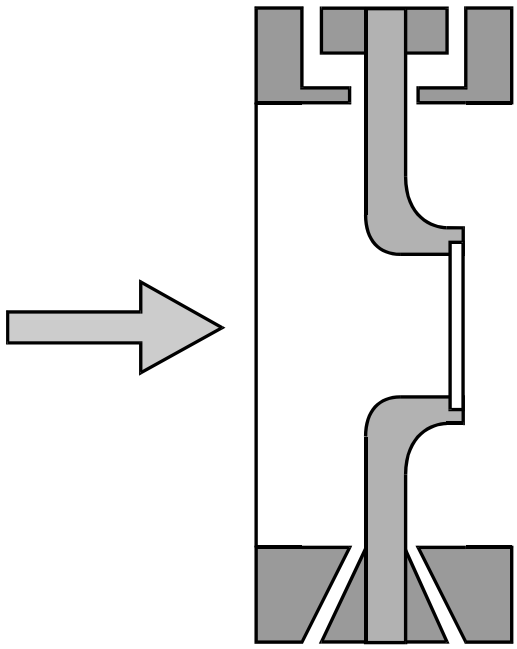


Figure 5.9
The flow nozzle is preferable for high velocity applications

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Advantages

- High velocity applications.
- Operate in higher turbulence.
- Used with fluids containing suspended solids.
- More cost effective than venturis.
- Physically smaller than the venturi.

Disadvantages

- More expensive than orifice plates.
- Higher unrecoverable pressure loss.

5.2.11 Flow Tube

Flow tubes are more compact than the venturi tube, but provide a higher differential pressure for less unrecoverable pressure loss. They are primarily a low loss meter and the most common type would be the Dall tube.

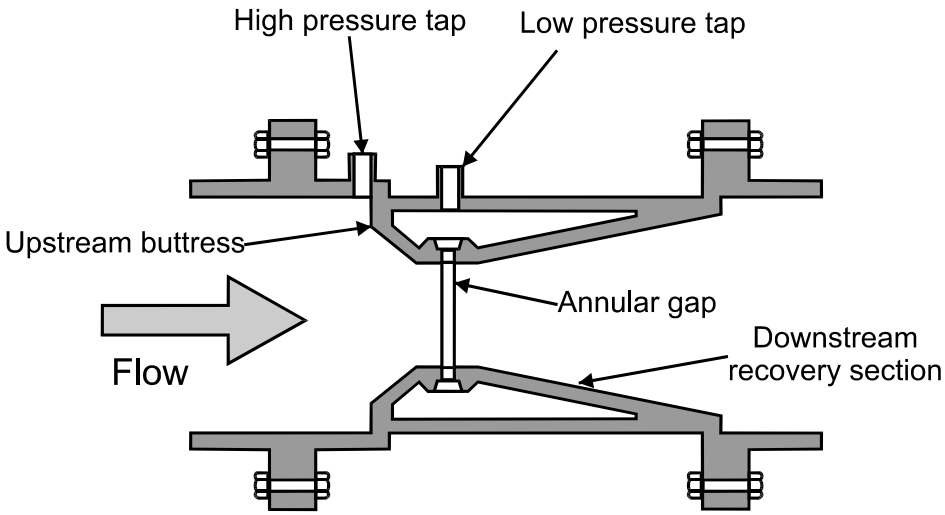


Figure 5.10
The Dall tube low-loss meter

Advantages

- Shorter lay length.
- Lower unrecoverable pressure loss.

Disadvantages

- More complex to manufacture.
- Sensitive to turbulence.
- Accuracy based on flow data.

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Application Limitations

It is recommended that when using this type of tube, it should be calibrated with the piping section in which it is to be used and over the full range of flows in which it will operate.

These types of flow devices do not necessarily conform to the wide range of conditions and extensive test data for orifice meters.

5.2.12 Pitot Tube

Refer to Chapter 2 for more information on this topic.

The Pitot tube measures flow based on differential pressure and is primarily used with gas flows.

The Pitot tube is a small tube that is directed into the flowstream. This measures the total pressure (dynamic and static combined). A second measurement is required, being of static pressure. The difference between the two measurements gives a value for dynamic pressure. The flowrate, like other devices, is calculated from the square root of the pressure.

In calculating the flowrate from the pressure, the calculation is dependent on such factors as tube design and the location of the static tap. The Pitot-static probe incorporates the static holes in the tube system to elminate this parameter.

The Pitot tube is also used to determine the velocity profile in a pipe. This is done by measuring points at various distances from the pipe wall to construct a velocity profile.

The Pitot tube is the primary device. It has the advantage over orifice meters of practically no pressure drop. Its usefulness is limited to clean gases and liquids as the sensing element is a small orifice. Foreign materials tend to plug the openings in the tube, and the classical Pitot tube senses impact pressure at one point only, thus decreasing accuracy.

Pitot tubes develop a very low differential pressure, which can often be difficult to measure with the secondary element. Also the accuracy of the device is dependant on the velocity profile of the fluid. The velocity profile is also affected by turbulence in the flow stream.

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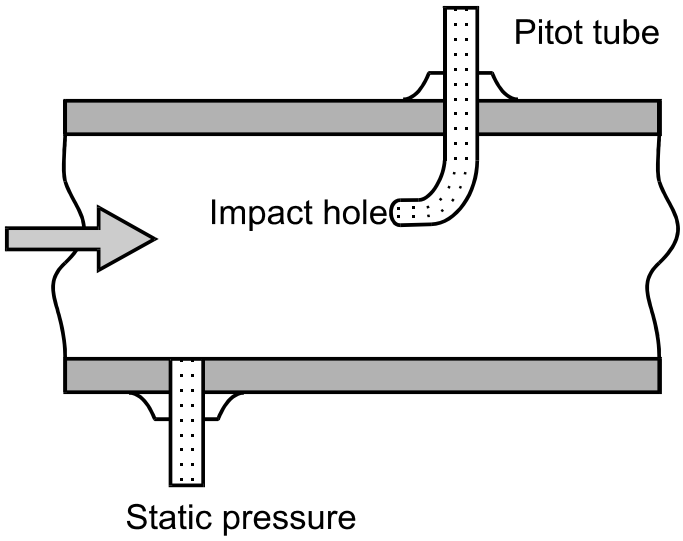


Figure 5.11
Principle of operation of the Pitot tube

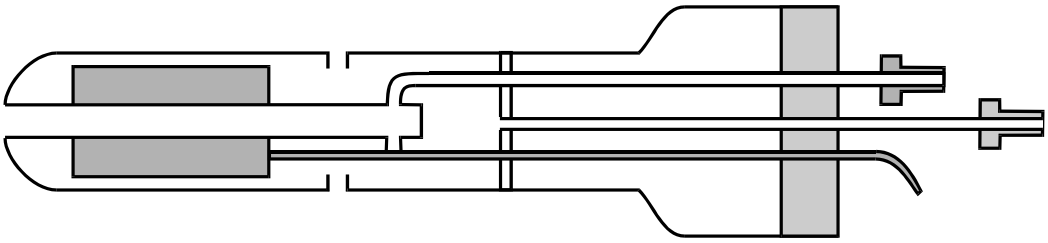


Figure 5.12
Basic form of pitot-static tube

Advantages

- Low cost.
- Low permanent pressure loss.
- Ease of installation into existing systems.

Disadvantages

- Low accuracy.
- Low Rangeability.
- Requires clean liquid, gas or vapour as holes are easily clogged.

5.2.13 Multiport Pitot Averaging

An annubar tube is a multi-impact opening type and improves the accuracy of this type of measurement. In most industrial applications this type of multi-impact or averaging is used to compensate for changes in the velocity profile.

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Annubar sensor are inserted perpendicular to the flowstream and extend the full diameter of the pipe. There is a very low obstruction to the flow, which causes minimal pressure loss.

Sensing ports are located on both upstream and downstream sides of the Annubar. These ports are connected to dual averaging plenums. The number of ports is proportional to the diameter of the pipe. The upstream ports produce an average impact (total) pressure and the downstream ports produce the average reference (static) pressure. The difference is an accurate and stable dynamic pressure that is easily converted into flowrate.

Annubars also provide good measurement when located in difficult piping. They can be located as close as two pipe diameters downstream of an elbow and still give accurate and repeatable measurements.

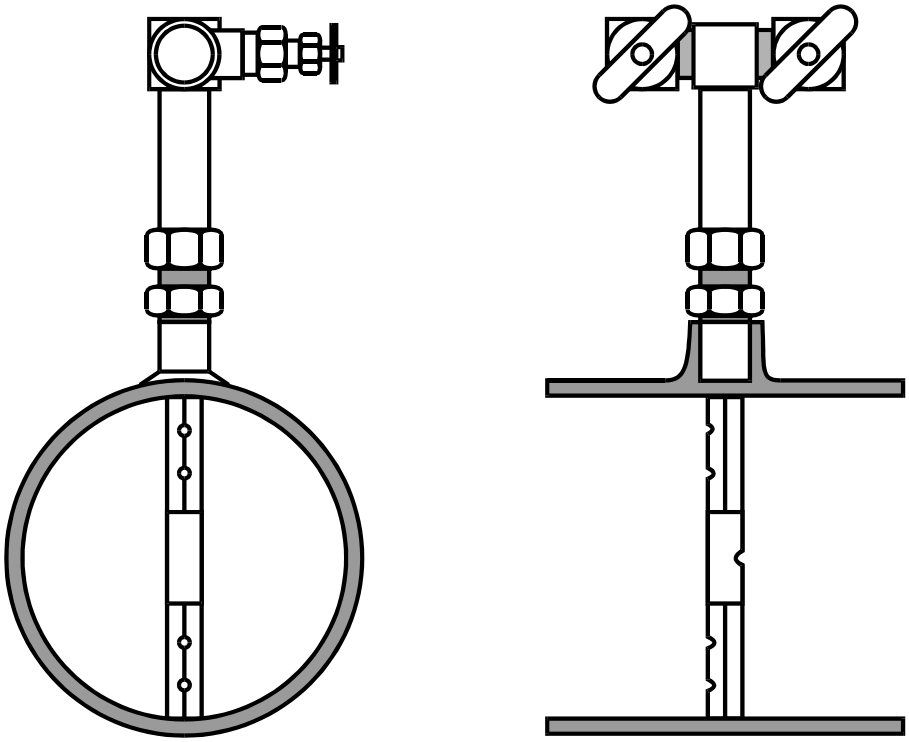


Figure 5.13
Multiport 'Annubar' Pitot averaging system
(courtesy of Dieterich Standard)

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5.2.14 Elbow

A pipe elbow can be used as a primary device. Elbow taps have an advantage in that most piping systems have elbows that can be used. In applications where cost is a factor and additional pressure loss from an orifice plate is not permitted, the elbow meter is a viable differential pressure device.

If an existing elbow is used then no additional pressure drop occurs and the expense involved is minimal. They can also be produced in-situ from an existing bend, and are typically formed by two tappings drilled at an angle of 45° through the bend. These tappings provide the high and low pressure tapping points respectively.

Tappings at 22.5° have shown to provide more stable and reliable readings and are less affected by upstream piping. However 45° tappings are more suited to bi-directional flow measurement.

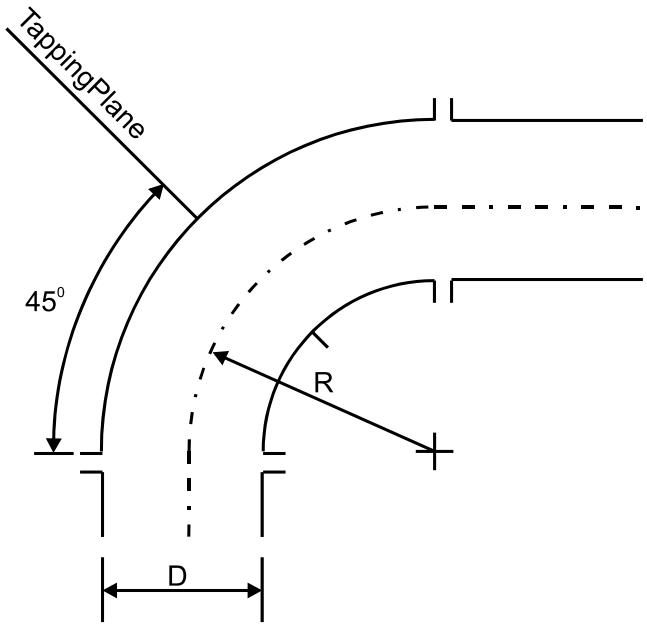


Figure 5.14
Elbow meter geometry

A number of factors contribute to the differential pressure that is produced. Because of the number of variables, it is difficult to accurately predict the exact flowrate. Some of these factors are:

- Force of the flow onto the outer tapping.
- Turbulence generated due to cross-axial flow due to the bend
- Differing velocities between outer and inner radius of flow
- Pipe texture
- Relationship between elbow radius and pipe diameter

Apart from these factors, accuracy from tests means that a predicted accuracy of less than 5% is possible. On site tests can be performed for more accurate results, with the added advantage that repeatability is good for this type of measurement.

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The disadvantages are that accuracy will be lacking (typically 5%) and dirty flows can plug the taps. At low-flow velocities, the differential produced is inadequate for good measurement, and is therefore only suitable for higher velocities.

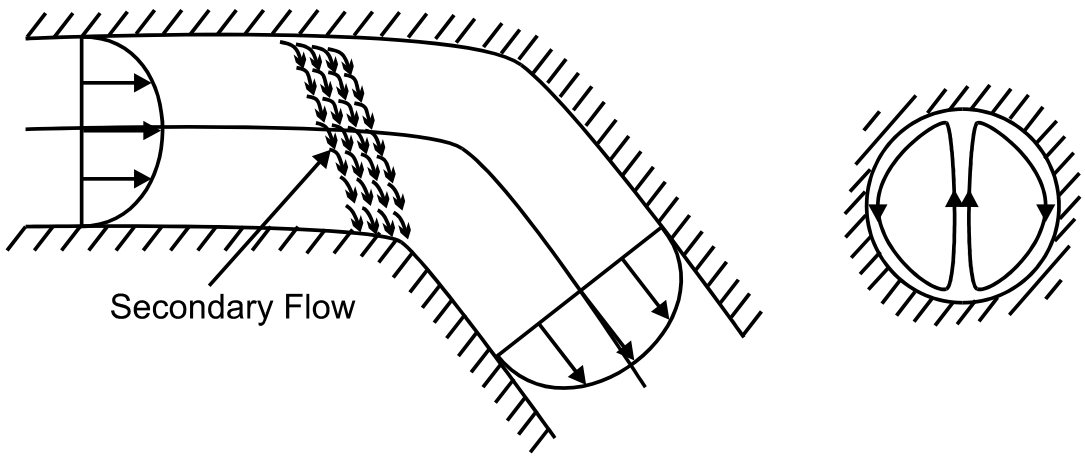


Figure 5.15
Flow in a pipe band

Although the elbow meter is not commonly used in industry, it is very much underrated. The low cost of the meter, together with its application after completion of pipework can be a major benefit for low accuracy flow metering applications.

Some suitable applications would include plant air conditioning, cooling water metering, site flow checkpoints possibly with local indicators, and check flow applications, where the cost of magmeters is prohibitive.

For installation, it is recommended that the elbow be installed with 25 pipe diameters of straight pipe upstream and at least 10 pipe diameters of straight pipe down stream.

Advantages

- Simplified installation.
- Inexpensive.

Disadvantages

- Low accuracy.

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5.2.15 Primary element - overview

Advantages

- No moving parts.
- Large range of sizes and opening ratios.
- Suitable for most gases and liquids.
- Well understood and proven.
- Price does not increase dramatically with size.

Disadvantages

- Accuracy is affected by density, pressure and viscosity fluctuations.
- Erosion and physical damage to the restriction affects measurement accuracy.
- Cause some unrecoverable pressure loss.
- Viscosity limits measuring range.
- Require straight pipe runs for expected accuracies.

Application Limitations

- Limited accuracy in measurement.
- Low Rangeability of 4:1.

Pressure loss:

When obstructing the flow using the differential pressure method of sensing, note that they do cause some unrecoverable pressure loss in the line.

Disturbances:

Differential type instruments that cause a restriction are easily affected by disturbances to inflow and outflow. Apart from bends in the pipe, such fittings as T-pieces and valves can cause disturbances to the flow measurement and as such should be separated from the instrument by a significantly long straight pipe extension. The length of clean pipe should be greater than 5 x diameter on the upstream side and no less than 4 x diameter on the downstream side.

5.2.16 Secondary Element

The secondary element is a differential pressure transmitter. This device provides the electrical output signal for interfacing to other instrumentation or control equipment. The output from this device is proportional to either the differential pressure or the flow rate.

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5.2.17 Troubleshooting

One of the most common inaccuracies induced in differential pressure flowmeters is not allowing enough straight pipe. When the flow material approaches and passes some change in the pipe small eddies are formed in the flow stream. These eddies are localised regions of high velocity and low pressure and can start to form upstream of the change and dissipate further downstream.

Flowmeter sensors detect these changes in pressure and consequently produce erratic or inaccurate readings for flow rate.

5.3 Open Channel Flow Measurement

The primary devices used in open channel flow measurements are weirs and flumes.

5.3.1 Weirs

Weirs are openings in the top of a dam or reservoir that allow for the flow of liquid, and enable measurement of the flow. With the characteristics of the weir known, the flow is generally determined by the height of the liquid in the weir.

There are two basic weirs - the rectangular and the V-notch.

Rectangular

There are three types of rectangular weirs:

1. Vertical walls with a flat base to form a boxlike opening in the channel
2. Open weir which extends the width of the channel
3. Cippoletti weir, has the end contractions set at a 4:1 angle

V-Notch

V-notch weirs are generally metal plates that contain a V-shaped notch.
The angle of the V can vary but the most common are for 30, 60 and 90 degrees.

V-notch weirs are used for lower flow rates than those that would be measured by a rectangular weir.

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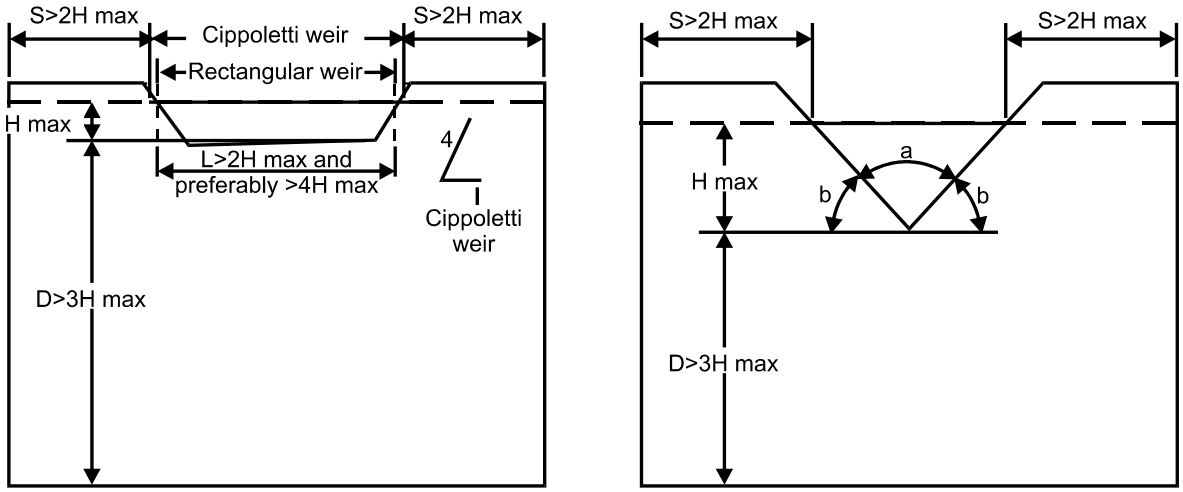


Figure 5.16
Rectangular, Cippoletti and V-notch weirs

Advantages

- Simple operation.
- Good Rangeability (for detecting high and low flow).

Disadvantages

- Pressure loss.
- Accuracy of about 2%.

Application Limitations

There is a high unrecoverable pressure loss with weirs, which may not be a problem in most applications. However with the operation of a weir, it is required that the flow clears the weir on departure. If the liquid is not free flowing and there is back pressure obstructing the free flow, then the level over the weir is affected and hence the level and flow measurement.

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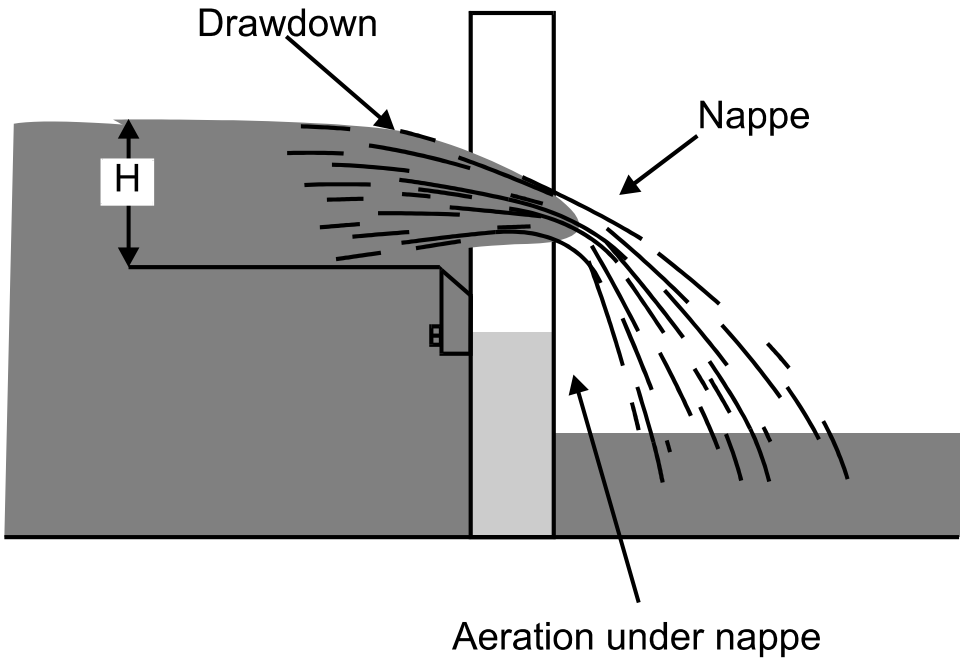


Figure 5.17
Flow over a weir

Summary

In weir measurement, the fluid must not go over the side of the weir if accuracy is to be maintained. All weirs produce some head loss as the liquid falls free. A flume is a better choice when head loss is a problem

5.3.2 Flumes

Flumes are a modification to the weir where the section of flow is reduced to maintain head pressure. A flume forces the liquid into a narrower channel and in doing so only incurs a head pressure drop of about 1/4 of that for a weir of equal size. This process is similar in principle to a rectangular venturi tube.

Similar to the weir, the water level in the flume is a function of flow rate. The flume provides a little more accuracy by precisely channelling the flow. The flume is also independent of the velocity of the fluid as it enters, and as such the application does not require damming or a stilling basin.

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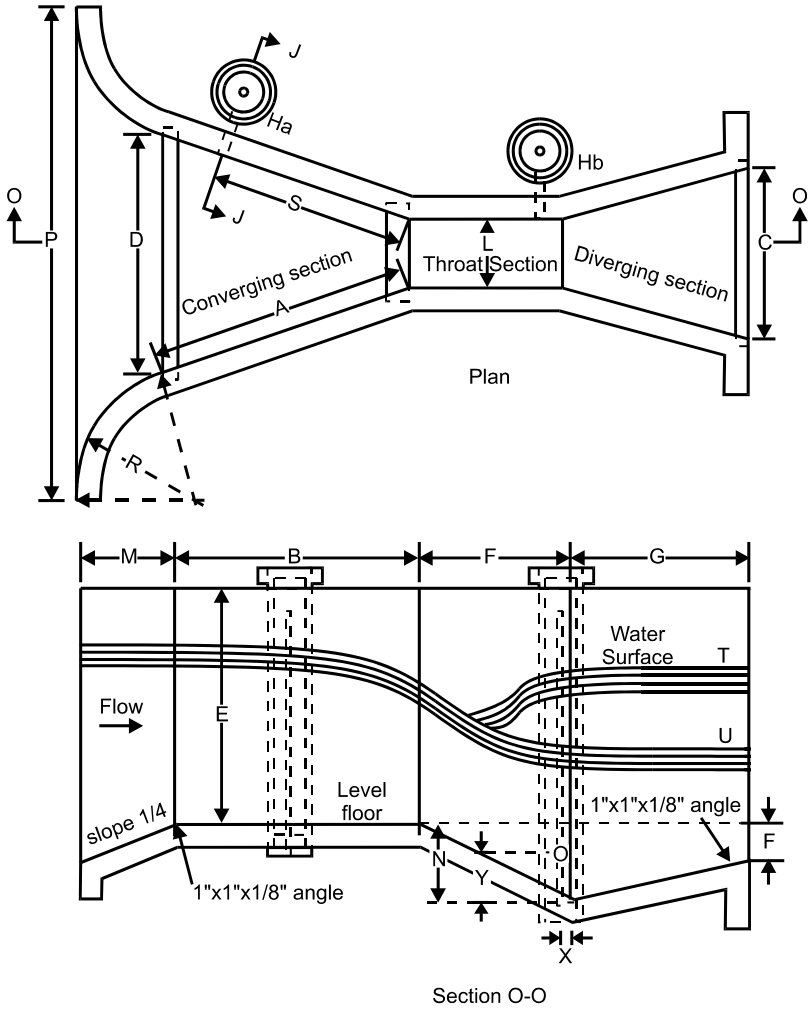


Figure 5.18
Diagram and dimensions of Parshall Flume

Typical Applications

Flumes are typically used for applications involving larger scale irrigation systems, water supply, waste management and effluent treatment plants.

Advantages

- Reliable and repeatable measurements.
- No erosion.
- Not sensitive to dirt and debris.
- Very low head pressure loss.
- Simple operation and maintenance.

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Disadvantages

- High installation costs.
- Low accuracy.
- Expensive electronics.

Application Limitations

Small flumes may be purchased and installed whereas larger flumes are generally fabricated on site.

Summary

The venturi flume has replaced the weir in most applications, and the Parshall flume is possibly the most accurate open channel flow measuring system at present.

Primary Element

Installation and Selection Considerations

Some of the factors to consider when selecting the primary element are:

- Probable flow range
- Acceptable head loss
- Accuracy and resolution
- Consistency of the fluid

This type of flow measurement will not function correctly if the weir or flume is overflowing. Minimum weir flows are also difficult to calculate because they depend on the nature of the weir. If the weir is sized too large then the readings will be negligible. A properly sized weir will produce a substantial, measurable amount of head over the weir for minimum flow.

Secondary Element

Measurement by weirs are very accurate under ideal conditions. However in the actual plant, measurement is affected by the velocity of approach factor and can have a typical error of 3 to 5 percent. The flow in an open channel system is related to the level and this is where the errors occur.

Also note that the equations for most open-channel primary devices are non-linear and extensive calculations would be required to convert head into flow rate. As such, the instrument best handles these calculations.

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Level Measurement

The head used for calculating the flow rate is measured using level sensing equipment. This is covered in Chapter 3, however some of the more common types for open channel flow measurement are:

- float and cable
- ball float
- air bubbles
- pressure sensing

In closed applications, and even in pipes, ultrasonics are often used to determine the level.

5.4 Variable Area Flowmeters

5.4.1 Principles of Operation

Variable Area flowmeters work with low viscous liquids at high velocities. The principle of operation is that the flow stream displaces a float placed in the stream. The rate of flow is related to the area produced by forcing the float up or down, and varying the area.

It is because of the low viscosity and high velocity that the frictional resistance of the flow is negligible compared to the resistance of the obstruction (float) placed in the flow stream.

The float in the early stages of development was slotted which caused the floats to rotate. This provided stability and centring of the float, and is where the designation of rotameter came from.

The rotameter consists of a tapered measuring tube and a float. This arrangement produces a resistance value (coefficient of resistance) for the float, which depends on its position in the measuring tube. A balance is achieved between the force of the flow stream and the weight of the float. The float positions itself vertically within the measuring tube such that the resistance value is balanced.

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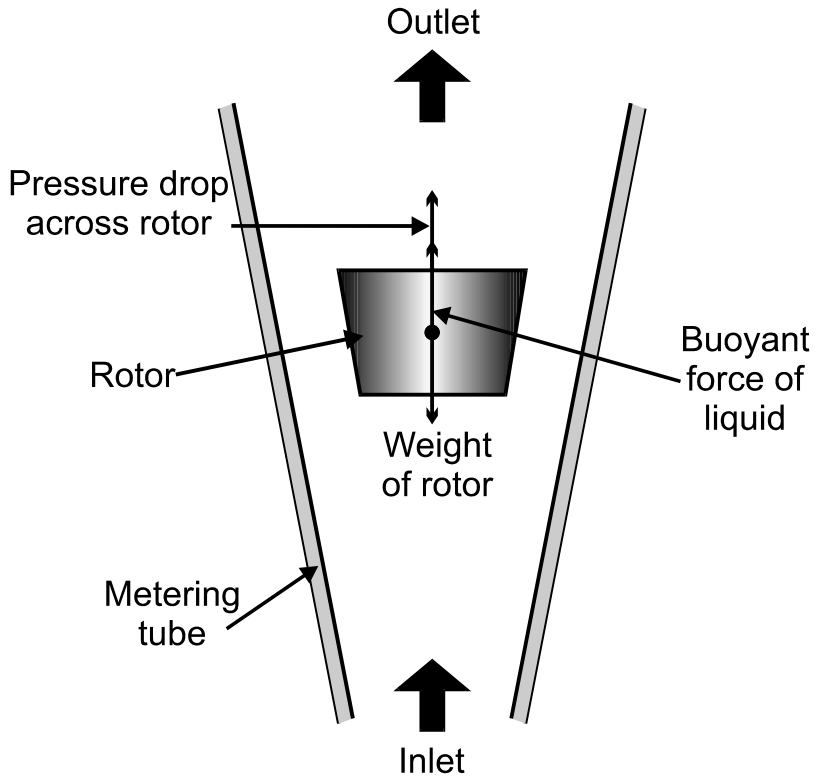


Figure 5.19
Variable area (Rotameter) flowmeter

The inside of the measuring tube is conical and has guide strips for the float. For physical indication, there is a scale on the outside to indicate the flowrate. Metal versions are available that have a means of transmitting the float position. The measuring tube can be made from steel, stainless steel, plastics (polypropylene, teflon), glass or hard rubber.

Also a number of various floats are available. The rotating float is used for direct control. Another type that is available is unaffected by viscosity, and a modification of this is available that magnifies the sensitivity of the operating range by 30%, but is more sensitive to viscosity.

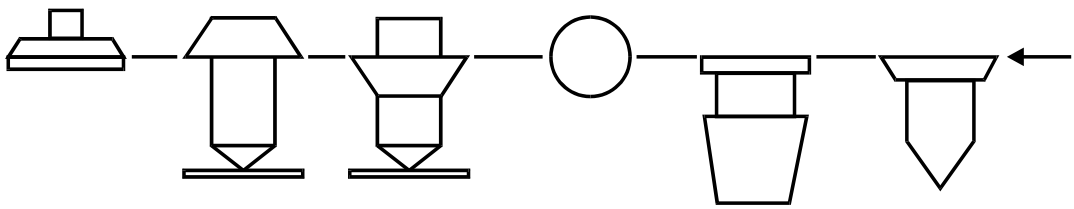


Figure 5.20
Viscosity limits of Rotameters depend on float shapes
(Courtesy of Brooks Instrument Division of Rosemount)

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5.4.2 Selection and Sizing

In sizing a rotameter, it is required that the actual flow be converted into a standard flow. Capacity tables are based on the standard flow, this is where the density of the float is taken into account. To assist in the sizing, manufacturers provide slide rules or comparison tables specifically designed for rotameter sizing.

The rotameter is not affected by piping configurations. It can be installed with any form of upstream piping prior to the meter.

For a 25mm bore the cost is comparable to magmeter.

5.4.3 Typical Applications

High accuracy rotameters are generally used in laboratory and pilot scale testing applications.

5.4.4 Advantages

- Inexpensive
- Wide range of applications
- Very basic operation
- Easy installation and simple to replace

5.4.5 Disadvantages

- Limited accuracy
- Subject to density, viscosity and temperature
- Fluid must be clean, no solids content
- Erosion of device (wear and tear)
- Can be expensive for large diameters
- Operate in vertical position only
- Viscosity > 200cP

5.4.6 Application Limitations

Glass tubes provide the added limitations of visibility and also impose limitations relating to pressure and temperature ratings. Metal metering tubes are used in applications where glass is not permissible. The float position is determined by magnetic or electrical means.

5.5 Oscillatory Flow Measurement

Oscillatory flow equipment generally measures the velocity of flow (hence they are often referred to as velocity flowmeters) and the volumetric flow rate is calculated from the relationship:

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$$Q = v \cdot A$$

where: Q is the volumetric flow rate
v is the velocity of the flow
A is the cross sectional area

The primary device generates a signal that is proportional to fluid velocity. This eliminates the errors that are amplified in square root calculations.

Velocity flowmeters, in general can operate over a greater range of velocities and are less sensitive to the flow profile compared with differential pressure devices.

5.5.1 Vortex Flowmeters

Vortex flowmeters can measure liquid, gas or steam and use the principle of vortex shedding. Vortex shedding occurs when an obstruction is placed in the flowing stream. The obstruction is referred to as a bluff body and causes the formation of swirls, called vortices, downstream from the body.

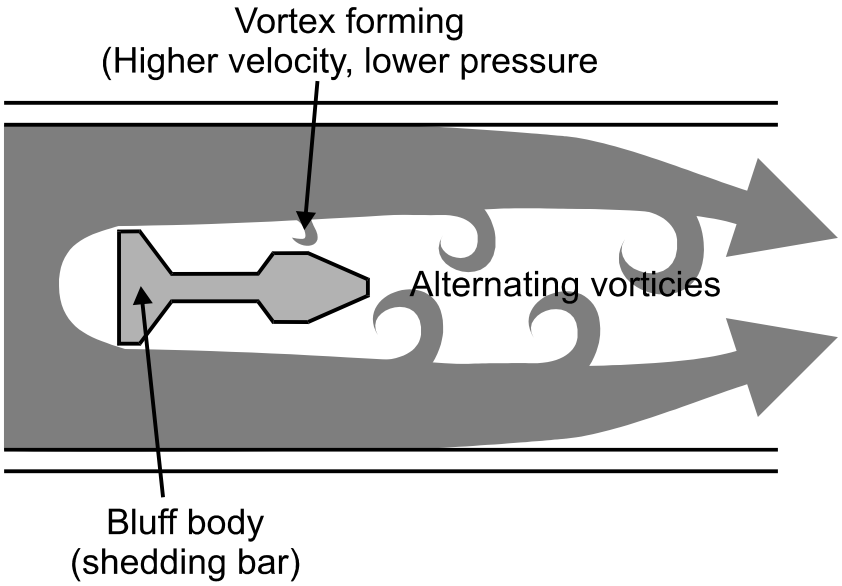


Figure 5.21
Principle of operation of vortex meter

Differential pressure changes occur as the vortices are formed and shed. This pressure variation is used to actuate the sealed sensor at a frequency proportional to the vortex shedding. For continuous flow, a series of vortices generates electrical pulses with a frequency that is also proportional to the flow velocity. The velocity can then be converted to volumetric flow rate.

The output of a vortex flowmeter depends on the K-factor. The K-factor relates to the frequency of generated vortices to the fluid velocity.

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$$Fluid\ velocity = \frac{Vortex\ Frequency}{K - Factor}$$

The K-factor varies with the Reynolds number, however it is virtually constant over a broad range of flows. Vortex flowmeters provide very linear flow rates when operated within the flat range.

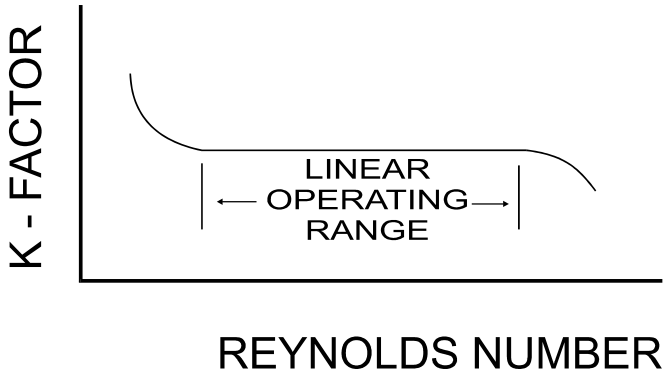


Figure 5.22
Relationship between K-Factor and Reynolds number

Primary Element - Bluff Bodies

Round Bluff Body

The initial bluff bodies were cylindrical and it was found that the separating point of the vortex fluctuated depending on the flow velocity. It was because of this movement that the frequency of the vortices was not exactly proportional to velocity.

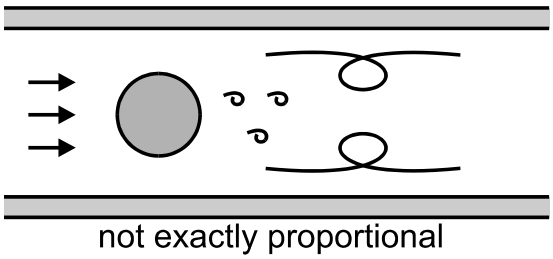


Figure 5.23
Round bluff bodies

Delta-Shaped Bluff Body

The Delta-shaped bluff body has been tested over many years and found to be ideal in terms of its linearity. Accuracy is not affected by pressure, viscosity or other fluid conditions. Many variations of the Delta shape exist and are in operation.

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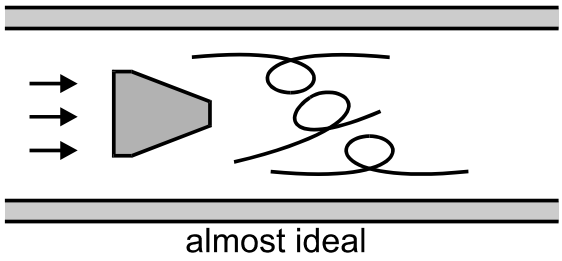


Figure 5.24
Delta shaped bluff bodies

Two-Part Bluff Body

Applications exist where a second bluff body is inserted behind the first. In this configuration, the first body is used to generate the vortices and the second body to measure them.

This method has the disadvantage of placing an extra obstruction in the stream. In such a case the pressure loss is almost doubled.

The advantage of this method is that a strong vortex is generated, which means that less complicated sensors and amplifiers may be used.

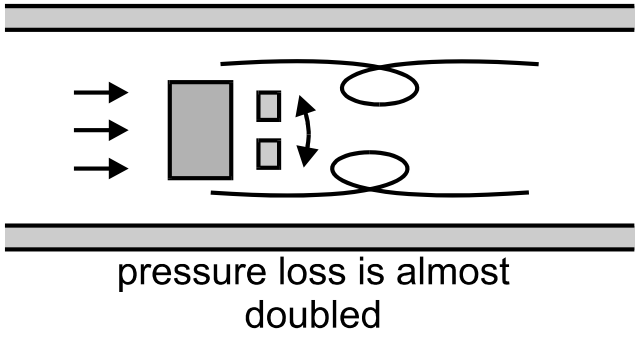


Figure 5.25
Two-part bluff body

Rectangular Bluff Bodies

One of the original bluff body designs was with a rectangular shape. This body style has been documented as unfavourable as they produces considerable fluctuations in linearity as the process density varies.

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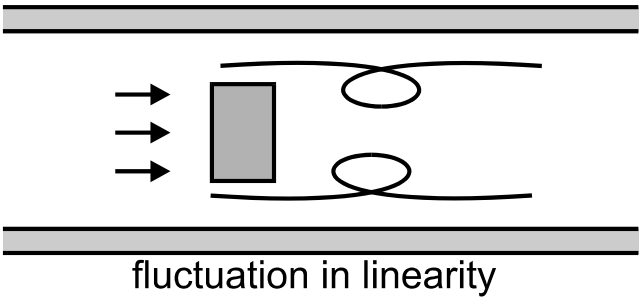


Figure 5.26
Rectangular bluff body

Vortex Swirl Type

A different type of vortex meter is that of the precession or swirl type. In this device, an internal vortex is forced into a helical path through the device. A thermistor is typically used to detect a change in temperature as the vortices pass. Again, the output signal is proportional to the flow rate.

A flow straightener is used at the outlet from the meter. This isolates the meter from any downstream piping effects that may affect the development of the vortex.

This type of measuring device has a Rangeability of about 10:1 and is used mainly with gases. Because of the higher tolerance in manufacture of this type of meter, it is more expensive then comparative meters.

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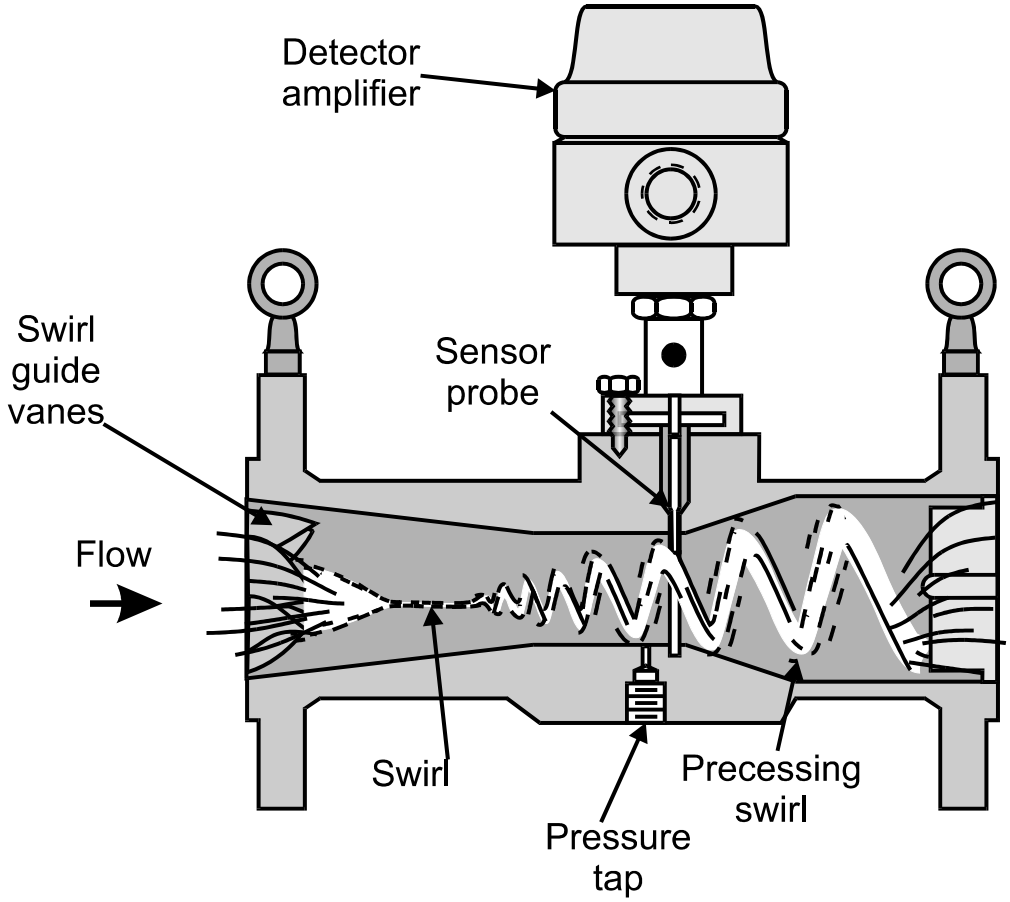


Figure 5.27
Construction of a typical vortex precession (swirl) meter

Coanda Flowmeter

The Coanda flowmeter generates internal oscillations similar to that of an electronic oscillator. The oscillations are generated by feeding part of the mainstream flow back into itself. There are two ports for feedback on opposing sides of the flowstream. The sensor (typically a thermistor) measures the frequency of oscillation, which is proportional to the rate of flow.

This flowmeter has a Rangeability of up to 30:1 and is used mainly with liquids.

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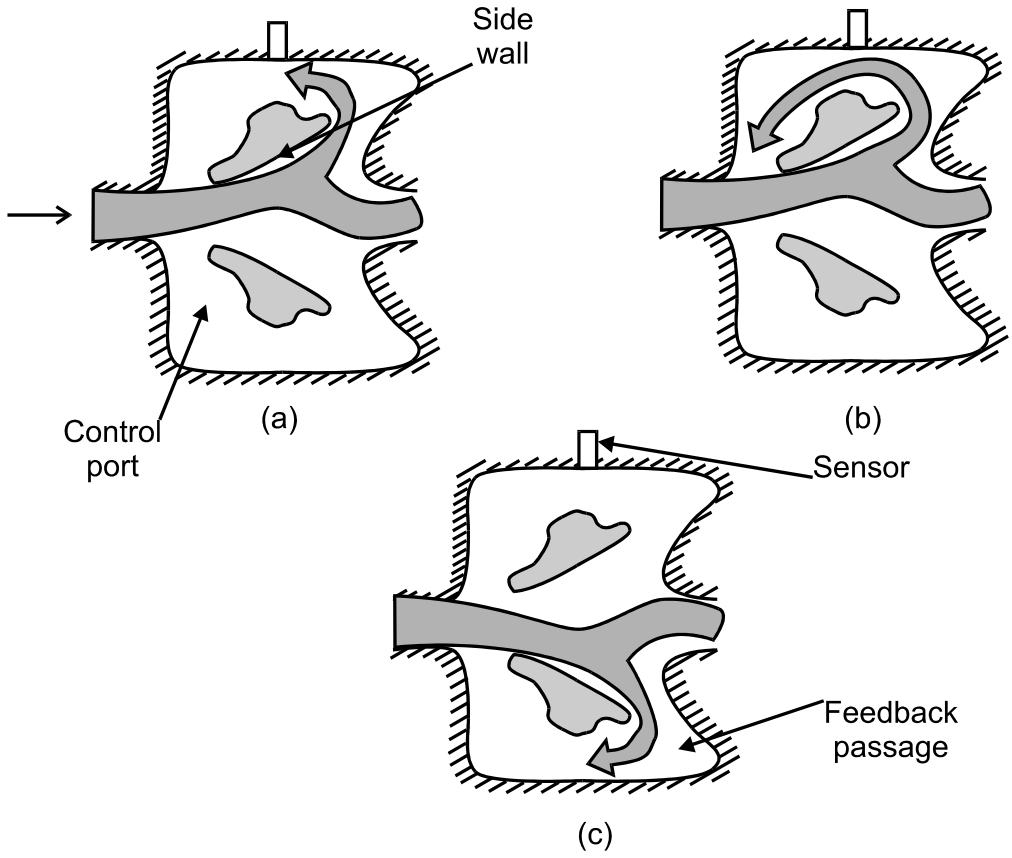


Figure 5.28
Diagram of the mode of operation of a feed back oscillator

Secondary Element

A number of devices can be used to measure the vortex frequency. The choice depends on the application, and more particularly the operating conditions.

- Thermistors
- Pressure sensors
- Magnetic pick-up
- Strain gauge
- Piezoelectric
- Capacitive

Thermistors

Thermistors are heat sensitive, and generate an ac voltage signal due to the cooling from the vortices. These are used in clean gas applications.

Pressure Sensors

As the vortices are generated and shed the pressure changes accordingly. A pressure sensor can be used where the pressure changes distort a diaphragm. A dielectric oil is used between the membrane and the insulated electrode and the change in

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capacitance is measured. Other devices use a piezoelectric crystal and measure a change in voltage. Temperature limitations apply due to the diaphragm and they are generally insensitive in the lower range. Suitable for fluids and possibly gas and low pressure steam.

Magnetic Pick-Up

A rather crude mechanical means of detecting the vortices is with a magnetic pick-up to sense a ball or disc which moves from side to side within the bluff body. Used for clean flows only as it can become blocked by dirt. Condensation problems can also occur when used in saturated steam, which can retard the action. Recommended for use with warm water, steam and low temperature liquids.

Strain Gauge

A strain gauge can be mounted on the bluff body to sense the pressure changes. Strain gauges work on a change in resistance and this can be used to measure the frequency of the vortices. Temperature limitations apply to the strain gauge itself. In addition vibration can be a problem for larger diameter pipes where a heavier bluff body is used. Strain gauges are excellent for gases and liquids.

Piezoelectric

A piezo device is a quartz crystal, which can be used in place of a strain gauge strip and can be mounted in the bluff body. The piezo device is more prone to failure, but is suitable for fluids, gases and steam.

Capacitive

Capacitive measurement has the added advantage of being immune to effects of pipe vibration.

Selection and Sizing

The following tables should assist with the selection and sizing of flowmeters.

Line sizes (mm)	Reynolds number limitations
12 - 100	10,000 minimum
150	20,000 minimum
200	35,000 minimum

Table 5.2
Minimum measurable Reynolds Numbers

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	Minimum (m/sec)	Maximum (m/sec)
Liquids	SQRT(54/ρ) or 0.22	SQRT(134,000/ρ) or 7.6
Gases (12 - 25mm)	SQRT(54/ρ) or 2.0	SQRT(134,000/ρ) or 67
Gases (40 - 200mm)	SQRT(54/ρ) or 2.0	SQRT(134,000/ρ) or 76

Table 5.3
Minimum and maximum measurable velocities

Line size, mm	K-factor, pulses/m^3
15	430,300
25	80,190
40	20,660
50	9,520
80	2,850
100	1,233
150	363.9
200	156.8

Table 5.4
Nominal flowmeter K-factors

Line size, mm	Minimum flow rate, m^3/hr	Maximum flow rate, m^3/hr
15	0.4	5.38
25	.61	15.3
40	1.09	35.9
50	1.81	59.4
80	4.00	130
100	6.86	225
150	23.6	512
200	54.5	885

Table 5.5
Minimum and maximum measurable flow rates - For Water at 25°C

Pressure Loss

The pressure loss varies depending on the manufacturers specifications. An example of the calculations required for determining the pressure loss for liquids and gases are provided below:

For Liquids:

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$$DP = \frac{(0.425) \times \rho_f \times Q_{lpm}^2}{D^4}$$

For Gases:

$$DP = \frac{(118) \times \rho_f \times Q_{acmh}^2}{D^4}$$

where:
- DP = Pressure loss, kPa
- ρ_f = Density at operating conditions, kg/m^3
- D = Flowmeter bore diameter, mm
- Q_{lpm} = Actual volumetric flow rate, l/min
- Q_{acmh} = Actual volumetric flow rate, m^3/hr

Minimum back pressure - liquids only

If there is too much of a pressure drop across the primary element, then cavitation can occur. By maintaining sufficient backpressure, this can be avoided. The minimum backpressure can be calculated as:

$$P = 2.9\ DP + 1.3\ \rho$$

where:
- P = Line pressure 5 diameters downstream of the meter
- DP = Pressure loss across the meter
- ρ_v = Liquid vapour pressure at operating conditions

Installation Techniques

Laminar flow is required for good measurement. Long straight lengths of pipe are generally used to produce a laminar flow profile.

If it is not possible to have such a section of pipe, the meter should be installed upstream of any disturbance. Where there is a severe upstream disturbance, the resulting long straight lengths of pipe can be reduced by fitting a radial vane or bundle-of-tubes in the upstream pipework. This is also termed a flow-straightening element that reduces turbulence and produces a more laminar flow.

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Advantages

- Suitable for liquid, gas or steam.
- Used with non-conductive fluids.
- No moving parts, low maintenance.
- Sensors available to measure both gas and liquid.
- Not affected by viscosity, density, pressure or temperature.
- Low installation cost.
- Good accuracy.
- Linear response .

Disadvantages

- Uni-directional measurement only.
- Clean fluids only.
- Not suitable with partial phase change.
- Not suitable for viscous liquids.
- Large unrecoverable pressure drop.
- Straight pipe runs required for installation.

Application Limitations

The upstream and downstream requirements of straight pipe vary according to a number of factors. A brief list of some of those factors is listed below:

- Severity and nature of disturbance upstream.
- Severity and nature of disturbance downstream.
- Type of votex bluff body (the specific design of the meter).
- The accuracy required.

The nature of the disturbance can be any of the following:

- Elbow
- Regulating control valve
- Reducer
- Expander

The straight upstream and downstream lengths required for the various installations in order to meet stated accuracies varies between manufacturers and products. Below is a figure that shows the pipe length requirements for installations as marketed by a particular supplier.

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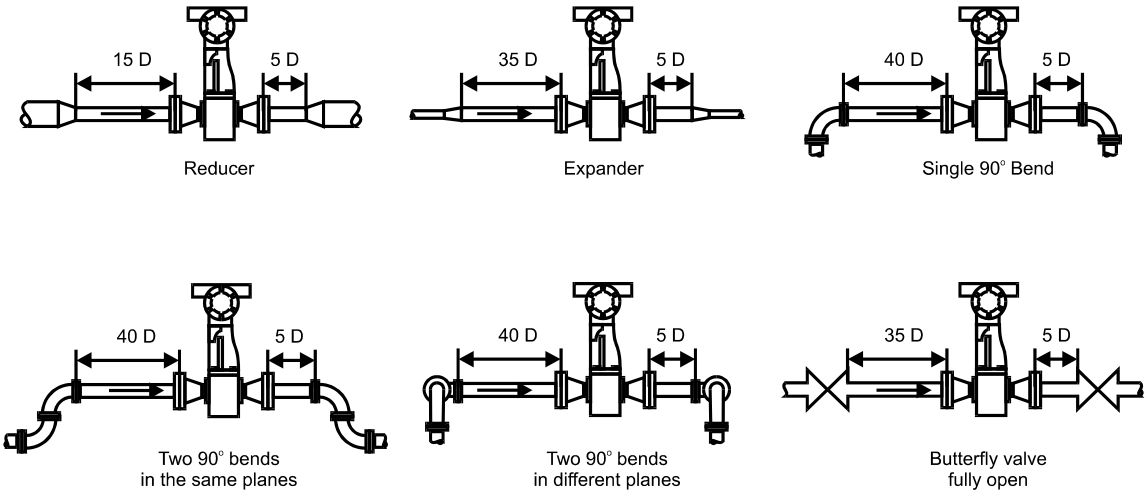


Figure 5.29
Straight pipe-run requirements as a function of upstream disturbance (Courtesy of Endress & Hauser)

Vortex flowmeters are not suitable where cavitation occurs and can be damaged, but will survive flashing conditions. Flashing conditions occur when some of the incoming liquid stream is permanently vaporised in the flowmeter.

Summary

Combinations of the same or different secondary elements are available that give slightly different specifications depending on the manufacturers products.

The vortex flowmeter appears to be the best answer for flow measurement of non-conductive fluids. Orifice plates and other differential pressure technology are being replaced with vortex flowmeters for measuring liquids, gases and steam. Although vortex meters are most widely used for steam measurement.

Vortex flowmeters generally have no moving parts and are unaffected by changes in temperature, pressure or density. They produce some unrecoverable pressure loss, but less than that for the same sized differential pressure flowmeter.

In general, the Rangeability is broad, and the meters can be used with liquids, gases or vapours simply by configuring the device.

Note: These devices work on the principle of generating vortices. Vortices are easily formed in high flow rates and low viscosities. Therefore these devices are not useful at very low flow rates, typically 0.3m/sec in liquid or 3m/sec in gases, or fluids with high viscosities. These devices produce a restriction in the line, and as such should not be used with erosive fluids, however they are more forgiving than orifice plates.

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5.5.2 Turbine

Basis of Operation

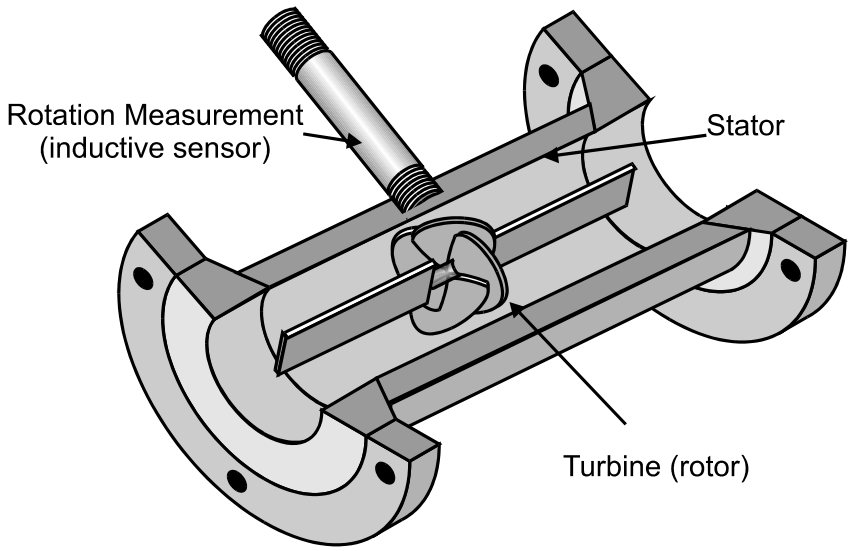


Figure 5.30
Turbine flowmeter

Turbine meters have rotor-mounted blades that rotate when a fluid pushes against them. They work on the reverse concept to a propeller system. Whereas in a propeller system, the propeller drives the flow, in this case the flow drives and rotates the propeller. Since it is no longer propelling the fluid, it is now called a turbine.

The rotational speed of the turbine is proportional to the velocity of the fluid.

Different methods are used to convey rotational speed information. The usual method is by electrical means where a magnetic pick-up or inductive proximity switch detects the rotor blades as they turn. As each blade tip on the rotor passes the coil it changes the flux and produces a pulse. Pulse rate is directly proportional to the flowrate.

As the rotation of the turbine is measured by means of non-contact, no tapping points are required in the pipe. Pressure is therefore not a problem, and in fact pressures of up to 9300psi can be applied without any problem, but this of course does depend on pipe diameter and materials of construction.

Temperature limitations are only imposed by the limitations of the materials of construction. To reduce losses or changes in process temperature, turbine flowmeters are available which can be subjected to wide temperature variations.

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Turbine meters require a good laminar flow. In fact 10 pipe diameters of straight line upstream and no less than 5 pipe diameters downstream from the meter are required. They are therefore not accurate with swirling flows.

They are not recommended for use with high viscosity fluids due to the high friction of the fluid which causes excessive losses as the turbine becomes too much of an obstruction. The viscosity of the liquid must be known for use of this type of meter.

They are also subject to erosion and damage. Each meter must be calibrated for its application.

Selection and Sizing

Turbine meters are sized by volumetric flow rate, however the main factor that affects the meter is viscosity.

Typically, larger meters are less affected by viscosity than smaller meters. This may indicate that larger meters would be preferred; in fact the opposite is true. By using a smaller meter, operation is more likely to occur towards the maximum permitted flowrate, and away from the non-linear response at low flows.

Turbine meters are specified with minimum and maximum linear flow rates that ensure the response is linear and the other specifications are met. For good Rangeability, it is recommended that the meter be sized such that the maximum flow rate of the application be about 70 to 80% of that of the meter.

Density changes have little effect on the meters' calibration.

K-Factor

The turbine meter measures volumetric flow, however the pulses produced vary depending on the meter. The variation is accounted for by a K-factor.

The K-factor is the number of pulses per unit volume. It is primarily determined by the size and type of the turbine meter. Due to manufacturing tolerances, the actual K-factor can vary between similar models.

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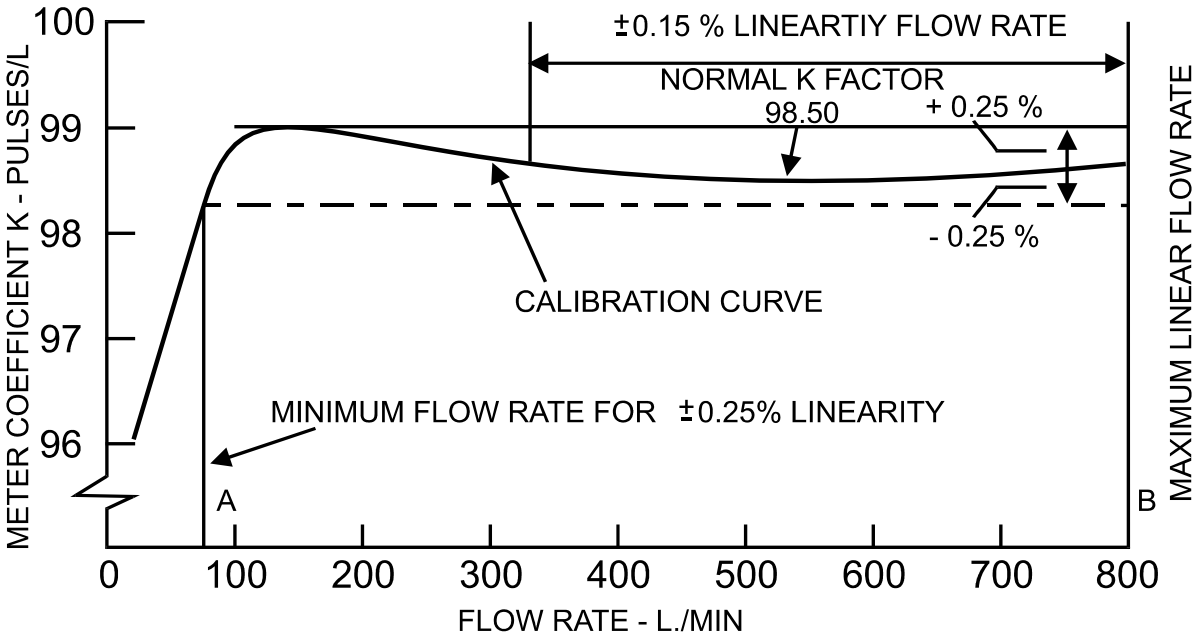


Figure 5.31
Typical calibration curve for a turbine meter

Advantages

- High accuracy, repeatability and Rangeability for a defined viscosity and measuring range.
- Temperature range of fluid measurement: -220°C to +350°C.
- Very high-pressure capability: 9300psi.
- Measurement of non-conductive liquids.
- Capability of heating measuring device..
- Suitable for very low flow rates.

Disadvantages

- Not suitable for high viscous fluids.
- Viscosity must be known.
- 10 diameter upstream and 5 diameter downstream of straight pipe is required.
- Not effective with swirling fluids.
- Only suitable for clean liquids and gases.
- Pipe system must not vibrate.
- Specifications critical for measuring range and viscosity.

Application Limitations

As turbine meters rely on the flow, they do absorb some pressure from the flow to propel the turbine. The pressure drop is typically around 20 to 30 kPa at the maximum flow rate and does vary depending on flow rate.

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It is a requirement in operating turbine meters that sufficient line pressure be maintained to prevent liquid cavitation. The minimum pressure occurs at the rotor, however the pressure recovers substantially are after the turbine.

If the backpressure is not sufficient, then it should be increased or a larger meter chosen to operate in a lower operating range. This does have the limitation of reducing the meter flow range and accuracy.

Summary

Turbine meters provide excellent accuracy, repeatability and rangeability for a defined viscosity and measuring range, and are commonly used for custody transfer applications of clean liquids and gases.

5.6 Magnetic Flowmeters

5.6.1 Basis of Operation

Electromagnetic flowmeters, also known as magmeters, use Faradays' law of electromagnetic induction to sense the velocity of fluid flow.

Faradays law states that moving a conductive material at right angles through a magnetic field induces a voltage proportional to the velocity of the conductive material. The conductive material in the case of a magmeter is the conductive fluid.

The fluid therefore must be electrically conductive, but not magnetic.

5.6.2 Power generators/magmeters - what's the connection?

If we consider power generators or tachometers that are used for speed measurement, then these operate on the same principle. Electrically conductive process fluid is fundamentally the same as the rotor in a generator. The fluid passes through a magnetic field induced by coils that are positioned around a section of pipe.

The process fluid is electrically insulated from the pipe with a suitable lining, in the case of a metal pipe, so that the generated voltage is not dissipated through the pipeline. The electrodes are located in the pipe and a voltage is generated across these electrodes that is directly proportional to the average velocity of the liquid passing through the magnetic field.

The coils are energised with ac power or pulsed dc voltage, so consequently the magnetic field and resultant induced voltage responds accordingly. The generated voltage is protected from interference, amplified and converted into a dc current signal by the transmitter. Line voltage variations are accounted for by the sensing circuits.

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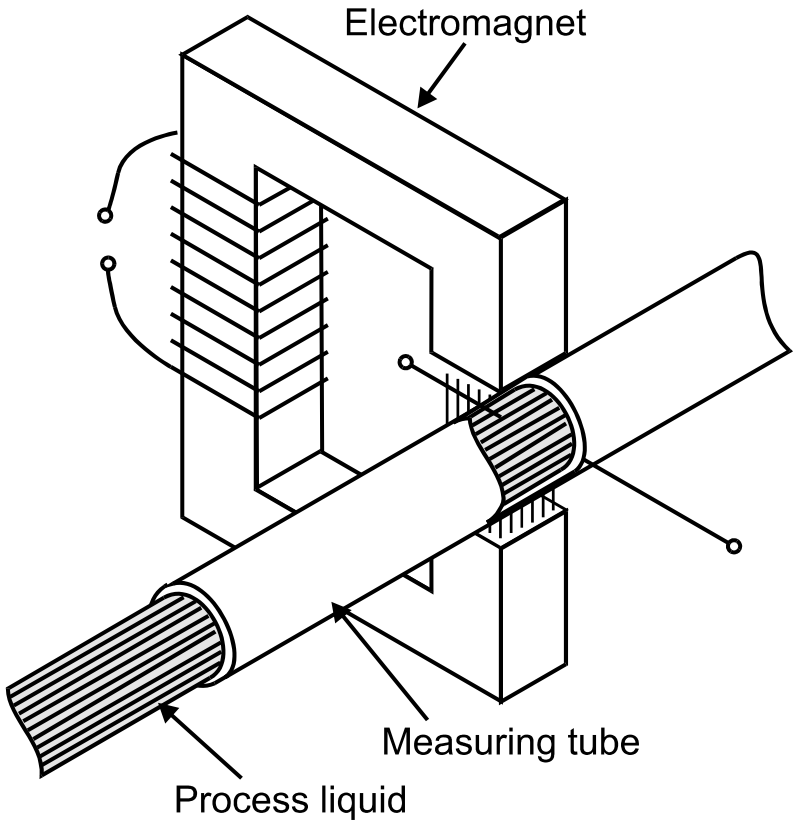


Figure 5.32
Principle of the electromagnetic meter

5.6.3 The Magmeter

The advantages of magnetic flowmeters are that they have no obstructions or restrictions to flow, and therefore no pressure drop and no moving parts to wear out. They can accommodate solids in suspension and have no pressure sensing points to block up. The magnetic flowmeter measures volume rate at the flowing temperature independent of the effects of viscosity, density, pressure or turbulence. Another advantage is that many magmeters are capable of measuring flow in either direction.

Most industrial liquids can be measured by magnetic flowmeters. These include acids, bases, water, and aqueous solutions. However, some exceptions are most organic chemicals and refinery products that have insufficient conductivity for measurement. Also pure substances, hydrocarbons and gases cannot be measured.

In general the pipeline must be full, although with the later models, level sensing takes this factor into account when calculating a flowrate.

Magnetic flowmeters are very accurate and have a linear relationship between the output and flowrate. Alternatively, the flow rate can be transmitted as a pulse per unit of volume or time.

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The accuracy of most magnetic flowmeter systems is 1% of fullscale measurement. This takes into account both the meter itself and the secondary instrument. Because of its linearity, the accuracy at low flow rates exceeds that of such devices as the Venturi tube. The magnetic flowmeter can be calibrated to an accuracy of 0.5% of full scale and is linear throughout.

5.6.4 Selection and Sizing - General

In most applications, the meter is sized to the pipe in which it is to be installed. Although this is probably the simplest approach, it is necessary to ensure that the meter meets the process requirements. The two main requirements are the maximum and minimum flows, although accuracy can be another deciding factor.

If these requirements cannot be achieved, then a meter of differing size to the pipeline is required. Under these conditions it is more common that the meter be downsized as the pipelines are often oversized.

Another reason for downsizing the meter is to increase the flowrate through the meter. In applications of liquids with a solids content, the flow velocity should be maintained above a certain level, typically 3 m/sec, to prevent build-up and deposits.

Sizing of magmeters is done from manufacturer's nomographs to determine suitable diameter meters for flow rates.

The conductivity of liquids can vary with temperature. Care should be taken to ensure the performance of the liquid in marginal conductivity applications is not affected by the operating temperatures. Most liquids do have a positive temperature coefficient of conductivity, however negative coefficients are possible in a few liquids.

Since the flowtube diameter determines the flow velocity, it becomes a very important consideration. The flowtube is sized to give a velocity between 0.3 and 9.1 m/sec.

Application	Velocity range, m/sec
Normal service	0.6 - 6.1
Abrasive slurries	0.9 - 3.1
Non-abrasive slurries	1.5 - 4.6

Table 5.6
Sizing guidelines for the type of application

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Calculating velocity:

Information is generally known about the required flow rate in the process, and from this the velocity needs to be determined. The manufacturer's specifications will give a Litres per minute factor for each size of meter.

$$Velocity = \frac{Flow\,rate}{Factor}$$

where: Velocity = speed of flow in m/sec
Flow rate = the known volumetric flow in l/min
Factor = obtained from the table below

Nominal size, mm	Litres per minute factor
15	11.834
25	33.657
40	79.297
50	130.67
80	287.88
100	495.73
150	1125.0
200	1948.3
250	3070.7
300	4437.2
350	5268.8
400	6881.5
450	8875.7
500	10826
600	15657
750	25036
900	36670

Table 5.7
Calculating flow rate

5.6.5 Selection and Sizing - Liners

The principle of operation of the magmeter requires the generation of a magnetic field, and the detection of the voltage across the flow. If the pipe is made of a material with magnetic properties, then this will disrupt the magnetic field and this effectively short circuits the magnetic field. Similarly if the inside of the pipe is conductive, then this will short circuit the electrodes used to detect the voltage across the flow.

The meter piping must be manufactured from a non-magnetic material such as stainless steel in order to prevent short-circuiting of the magnetic field.

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The lining of the meter piping must also be lined with an insulating material to prevent short-circuiting of the electric field.

The liner has to be chosen to suit the application, particularly the resistance it has to the following:

- chemical corrosion
- erosion
- abrasion
- pressure
- temperature

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Commonly used magnetic flowmeter liner materials					
Material	**General**	**Corrosion resistance**	**Abrasion resistance**	**Temperature limit (°C)**	**Pressure limit (bar)**
Teflon PFA (PFTE)	Warm-deformable resin with excellent ant-stick properties and suitable for food and beverage	Excellent	Fair	220	40
Teflon PFA	Melt-processable resin with better shape accuracy, abrasion resistance and vacuum strength than PTFE	Excellent	Good	180	40
Polyurethane	Extreme resistance to wear and erosion but not suitable for strong acids or bases	Wide range	Excellent	40	250
Neoprene	Combines some of the resistance to chemical attach of PTFE with a good degree of abrasion resistance	Wide range	Good to excellent	80	100
Hard rubber	Inexpensive – finds its main application in the water and waste water industries	Fair to excellent	Fair	90	250
Soft rubber	Mainly used for slurries	Fair	Excellent	40	64
Fused aluminium oxide	Highly recommended for very abrasive and/or	Excellent	Excellent	180	40

Table 5.8
Commonly used magnetic flowmeter liner material

A Summary of the characteristics of the various lines are:

Teflon (PTFE):

- Widely used due to its high temperature rating.
- Anti-stick properties reduce problems with build-up.
- Approved for food and beverage environments.
- Resistant to many acids and bases.

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Neoprene

- Good abrasion resistance.
- Good chemical resistance.

Soft Rubber

- Relatively inexpensive.
- High resistance to abrasion.
- Used mainly for slurry applications.

Hard rubber

- Inexpensive.
- General purpose applications.
- Used mainly for water and soft slurries.

Ceramic

- High abrasion resistance.
- High corrosion resistance.
- High temperature rating.
- Less expensive to manufacture.
- Also suited to sanitary applications.
- Strong compressive strength, but poor tensile strength.
- Brittle.
- May crack with sudden temperature changes, especially downward.
- Cannot be used with oxidising acids or hot concentrated caustic.

5.6.6 Installation Techniques

For correct operation of the magmeter, the pipeline must be full. This is generally done by maintaining sufficient backpressure from downstream piping and equipment. Meters are available that make allowances for this problem, but are more expensive and are specialised. This is mainly a problem in gravity feed systems.

Magmeters are not greatly affected by the profile of the flow, and are not affected by viscosity or the consistency of the liquid. It is however recommended that the meter be installed with 5 diameters of straight pipe upstream and 3 diameters of straight pipe downstream from the meter.

Applications requiring reduction in the pipe diameter for the meter installation, need to allow for the extra length of reducing pipe. It is also recommended that in those applications that the reducing angle not be greater than 8°, although manufacturers' data should be sought.

Grounding is another important aspect when installing magmeters, and manufacturers' recommendations should be adhered to. Such recommendations

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would require the use of copper braid between the meter flange and pipe flange at both ends of the meter. These connections provide a path for stray currents and should also be grounded to a suitable grounding point. Magmeters with built in grounding electrodes eliminate this problem, as the grounding electrode is connected to the supply ground.

5.6.7 Typical Applications

Magmeters are used in many applications as most liquids and slurries are suitable conductors. They are also the first to be considered in corrosive and abrasive applications. They can also be used for very low flow rates and small pipe diameters.

5.6.8 Advantages

- No restrictions to flow.
- No pressure loss.
- No moving parts.
- Good resistance to erosion.
- Independent of viscosity, density, pressure and turbulence.
- Good accuracy.
- Bi-directional.
- Large range of flow rates and diameters.

5.6.9 Disadvantages

- Expensive.
- Most require a full pipeline.
- Limited to conductive liquids.

5.6.10 Application Limitations

As mentioned earlier, a magnetic flowmeter consists of either a lined metal tube, usually stainless steel because of its magnetic properties, or an unlined non-metallic tube. The problem can arise if the insulating liners and electrodes of the magnetic flowmeter become coated with conductive residues deposited by the flowing fluid. Erroneous voltages can be sensed if the lining becomes conductive.

Maintaining high flow rates reduces the chances of this happening. However, some manufacturers do provide magmeters with built in electrode cleaners.

Block valves are used on either side of ac-type magmeters to produce zero flow and maintain a full pipe to periodically check the zero calibration limit. Dc units do not have this requirement.

5.6.11 Summary

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Magmeters are available with a ceramic lining and capable of measuring ultra-low conductivity fluids down to 0.01 uS/cm. Because of their electrode-less construction they are virtually immune to the effects of build-up and coating on the flowmeter liner.

5.7 Positive Displacement

Positive displacement meters measure flow rate by repeatedly passing a known quantity of fluid from the high to low pressure side of the device in a pipe. The number of times the known quantity is passed gives information about the totalised flow. The rate at which it is passed is the volumetric flow rate. Because they pass a known quantity, they are ideal for certain fluid batch, blending and custody transfer applications. They give very accurate information and are generally used for production and accounting purposes.

5.7.1 Rotary vane

Spring loaded vanes slide in and out of a channel in a rotor so that they make constant contact with the eccentric cylinder wall. When the rotor turns, a known volume of fluid is trapped between the two vanes and the outer wall. The flow rate is based on volume per revolution.

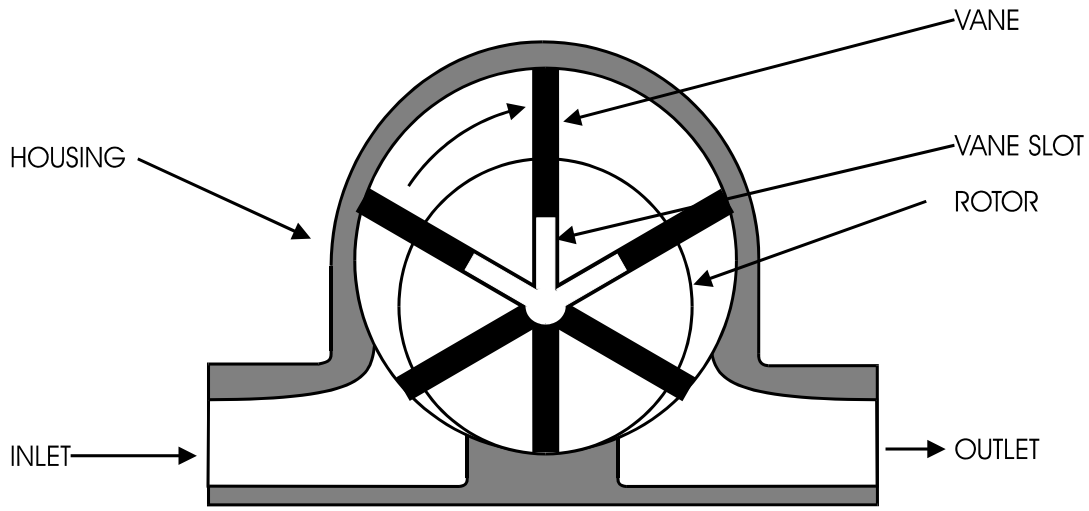


Figure 5.33
Rotating vane meter

The piston type is suitable for accurately measuring small volumes and is not affected by viscosity. Limitations with this device are due to leakage and pressure loss.

Typical Applications

This type of meter is used extensively in the petroleum industry for such liquids as gasoline and crude oil metering.

Advantages

- Reasonable accuracy of 0.1%.
- Suitable for high temperature service, up to 180°C.
- Pressures up to 7Mpa.

Disadvantages

- Suitable for clean liquids only.

Lobed impeller

This type of meter uses two lobed impellers, which are geared and meshed to rotate at opposite directions within the enclosure. A known volume of fluid is transferred for each revolution.

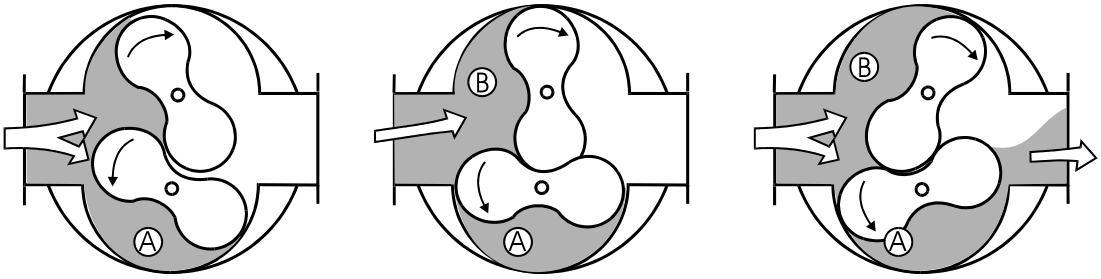


Figure 5.34
Rotating lobe meter

Typical Applications

The lobed impeller meter is often used with gases.

Advantages

- High operating pressures, up to 8Mpa.
- High temperatures, up to 200°C.

Disadvantages

- Poor accuracy at low flow rates.
- Bulky and heavy.
- Expensive.

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5.7.2 Oval gear meters

Two oval gears are intermeshed and trap fluid between themselves and the outer walls of the device. The oval gears rotate due to the pressure from the fluid and a count of revolutions determines the volume of fluid moving through the device.

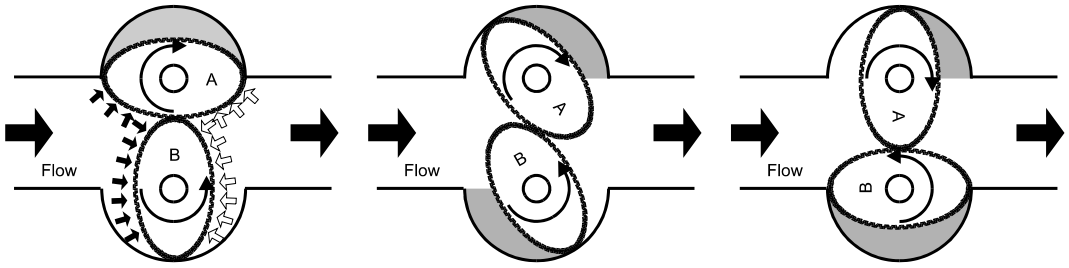


Figure 5.35
Positive displacement meters measure volumetric flow rate directly by dividing a stream into distinct segments of known volume, counting segments and multiplying by the volume of each segment.

The viscosity of the fluid can affect the leakage, or slip flow. If the meter is calibrated on a particular fluid, it will read marginally higher should the viscosity rise.

Newer designs of this type of meter use servomotors to drive the gears. This eliminates the pressure drop across the meter and also the force required to drive the gear. This eliminates the force, which causes the slip flow. This mainly applies to smaller sized meters and significantly increases the accuracy at low flows.

Advantages

- High accuracy of 0.25%
- High operating pressures, up to 10MPa
- High temperatures, up to 300°C
- Wide range of materials of construction

Disadvantages

- Pulsations caused by alternate drive action

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5.7.3 General Summary

Wear and damage is a problem due to the amount of contact between the parts and as such can only be used with clean fluids. The working life of the device also depends on the fluid being measured, this relates to solids build-up and temperatures of the fluid.

They are an obstruction to the flow path and consequently cause some pressure loss. Errors can occur due to leakage around the gears or pistons, but are reduced by using fluids that have some resistance to flow. Such viscous fluids have the ability to seal the small clearances. However if the fluid is too viscous then it can coat the inner chambers of the device and reduce the volume passed thus causing further errors.

Liquid positive displacement meters are one of the most common and widely used devices for measuring volumetric flow for custody transfer. They are simple and easy to maintain by regular maintenance personnel.

5.7.4 Application Limitations

Overspeeding can damage positive displacement meters. When the outlet pressure needs to be maintained due to an unacceptable pressure drop across the meter, it is quite common for the inlet pressure to be increased.

Specifications do vary, but in general they are primarily suited for clean, lubricating and non-abrasive applications.

Filters may be required to filter debris and clean the fluid before the meter. Regular maintenance is an obvious addition in this case. The added pressure drop may also need to be considered, especially if regular maintenance is not carried out.

Limitations on operating temperature can prove to be an inhibiting factor. If leakage does occur and is calibrated for, it can change with temperature as the viscosity varies.

One of the main limitations with this form of flow measurement is that the meter is driven by the flow. Particularly in the case of the oval gear meter, the force required to rotate the gear action is not constant and results in pulsations. These pulsations may may make the use of this type of meter impossible, particularly in controlled applications requiring a steady flow.

Positive displacement meters become limited when high volume measurement is required. They are primarily used for low volume applications.

In comparison, they are more expensive than magnetic flowmeters, but do have the added advantage of being able to measure non-conductive fluids.

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5.7.5 Advantages

- Can measure non-conductive fluids.
- Very high accuracy.
- Unaffected by viscosity.
- High Rangeability of up to 10:1.

5.7.6 Disadvantages

- Clean fluids only, limited life due to wear.
- Some unrecoverable pressure loss.
- Requires viscous fluid, not suitable for gas.
- Limited operating range.
- Mechanical failure likely to cause blockage in pipe.
- Cost.

5.8 Ultrasonic Flow Measurement

There are two types of ultrasonic flow measurement:

- Transit time measurement
- Doppler effect

The fundamental difference is that the transit-time method should be used for clean fluids, while the Doppler reflection type used for dirty, slurry type flows.

5.8.1 Transit time

The transit-time flowmeter device sends pulses of ultrasonic energy diagonally across the pipe. The transit-time is measured from when the transmitter sends the pulse to when the receiver detects the pulse.

Each location contains a transmitter and receiver. The pulses are sent alternatively upstream and downstream and the velocity of the flow is calculated from the time difference between the two directions.

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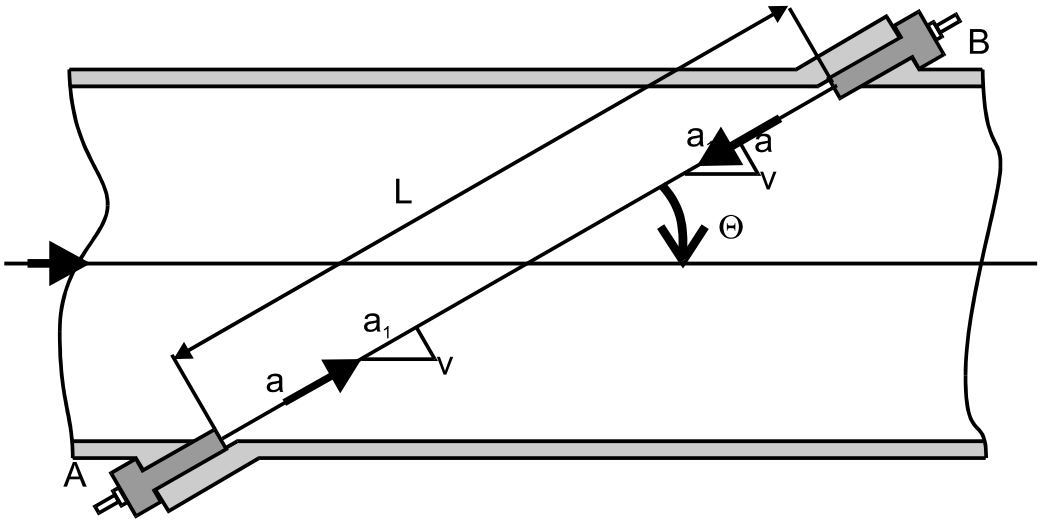


Figure 5.36
Transit time measurement

5.8.2 Installation Techniques

Designs are available that allow installation and removal of the transducers without interrupting the process flow. However there are three main options available:

- Fitted section of pipe
- Clamp on
- Transducers installed in-situ

The first option is where the manufacturer supplies a section of pipe fitted with the transducers factory mounted. These units have the advantage of being calibrated by the manufacturer to meet specifications. Allowances need to be made when fitting this section of pipe - something that can complicate existing installations.

Clamp-on transducers have the added advantage of being easy to install. They are mounted outside of the existing pipe. Since no section of pipe need to be installed then this type of flowmeter is easily retrofitted onto an existing system. They can be installed on metal, plastic and ceramic pipes.

Because they are portable and non-intrusive, clamp-on devices provide a good means of determining flowrates of unknown flows in existing installations.

A cheaper option is to install the transducers into the pipework. This does require tapping into the pipe and care needs to be taken to ensure the correct angles and tolerances are adhered to. This method generally requires calibration by the user once installed.

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5.8.3 Application Limitations

Clamp-on designs are limited because of the differing mediums in which the ultrasonics signals pass through. For optimum results, a sound-conductive path is required between the transducer and the process fluid inside the pipe. Couplings are available for reducing these effects but are quite expensive.

5.8.4 Typical Applications

Transit-time ultrasonic flow measurement is suited for clean fluids. Some of the more common process fluids consist of water, liquefied gases and natural gas.

5.8.5 Doppler Effect

The Doppler effect device relies on objects with varying density in the flowstream to return the ultrasonic energy. With the Doppler effect meter, a beam of ultrasonic energy is transmitted diagonally through the pipe. Portions of this ultrasonic energy are reflected back from particles in the stream of varying density. Since the objects are moving, the reflected ultrasonic energy has a different frequency. The amount of difference between the original and returned signals is proportional to the flow velocity.

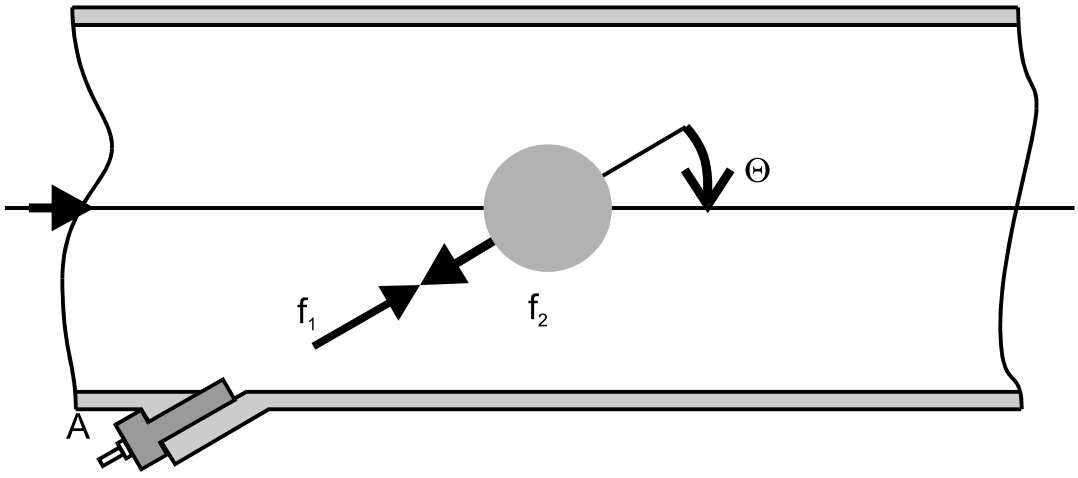


Figure 5.37
Doppler effect

It is quite common for only one sensor to be used. This contains both the transmitter and receiver. These can also be mounted outside of the pipe.

5.8.6 Application Limitations

As the Doppler flowmeter relies on reflections from the flowstream, it therefore requires sufficiently sized solids or bubbles. It is also required that the flow be fast enough to maintain a suitable quantity of solids and bubbles in suspension.

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Higher frequency Doppler flowmeters are available, but are limited to cleaner fluids.

5.8.7 Summary - General

Most ultrasonic flowmeters are mounted on the outside of the pipe and thus operate without coming in contact with the fluid. Apart from not obstructing the flow, they are not affected by corrosion, erosion or viscosity. Most ultrasonic flowmeters are bi-directional, and sense flow in either direction.

5.8.8 Advantages

- Suitable for large diameter pipes.
- No obstructions, no pressure loss.
- No moving parts, long operating life.
- Fast response.
- Installed on existing installations.
- Not affected by fluid properties.

5.8.9 Disadvantages

- Accuracy is dependent on flow profile.
- Fluid must be acoustically transparent.
- Errors cause by build up in pipe.
- Only possible in limited applications.
- Expensive.
- Pipeline must be full.

5.8.10 Application Limitations

Turbulence or even the swirling of the process fluid can affect the ultrasonic signals. In typical applications the flow needs to be stable to achieve good flow measurement, and typically allowing sufficient straight pipe up and downstream of the transducers does this. The straight section of pipe upstream would need to be 10 to 20 pipe diameters with a downstream requirement of 5 pipe diameters.

For the transit time meter, the ultrasonic signal is required to traverse across the flow, therefore the liquid must be relatively free of solids and air bubbles. Anything of a different density (higher or lower) than the process fluid will affect the ultrasonic signal.

5.8.11 Summary

Doppler flowmeters are not high accuracy or high performance devices, but do offer an inexpensive form of flow monitoring. Their intended operation is for dirty fluids, and find applications in sewage, sludge and waste water processes.

Being dependent on sound characteristics, ultrasonic devices are dependent on the flow profile, and are also affected by temperature and density changes.

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5.9 Mass Flow Meters

Mass flow measurement gives a more accurate account of fluids, and is not affected by density, pressure and temperature (unlike volumetric measurements).

Although most meters can infer mass flow rate from volumetric flow measurements, there are a number of ways to measure mass flow directly:

- The Coriolis meter
- Thermal mass flowmeter
- Radiation density

5.9.1 The Coriolis Meter

The Coriolis Effect

The basis of the Coriolis meter is Newtons' Second Law of Motion, where:

Force = Mass x Acceleration.

The conventional way to measure the mass of an object is to weigh it. In weighing, the force is measured with a known acceleration (9.81m/sec^2). This type of measuring principle is not easy or possible with fluids in motion, particularly in a pipe.

However, it is possible to manipulate the above formula and apply a known force and measure, instead, the acceleration to determine the mass.

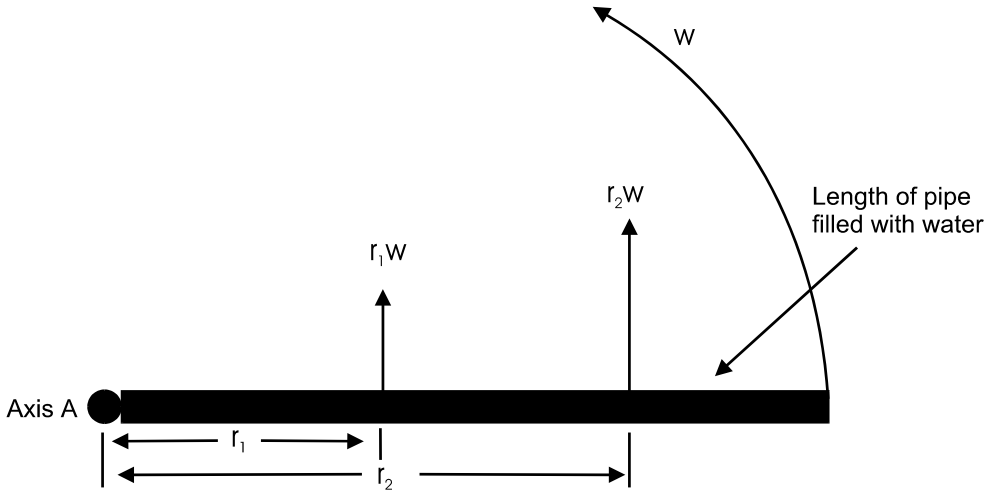


Figure 5.38
Principle of Coriolis effect

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The Coriolis effect causes a retarding force on a rotating section of pipe when flow is moving outward, conversely producing an advance on the section of pipe for flow moving towards the axis of rotation.

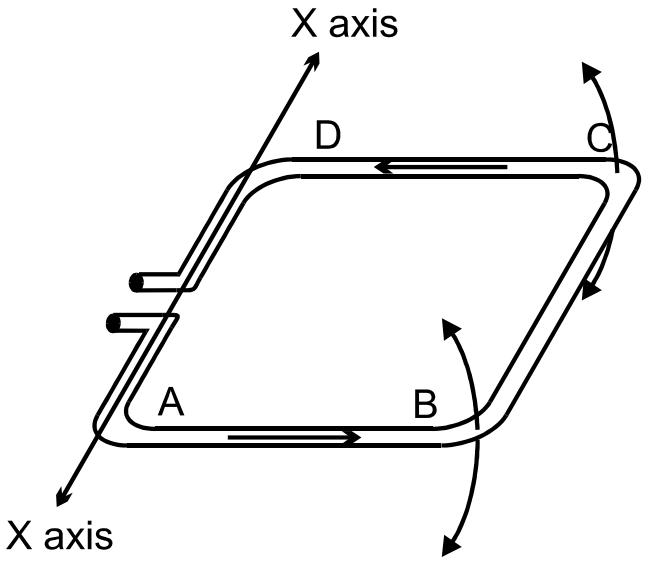


Figure 5.39
Application of Coriolis force to a meter

When the full section of pipe is moved about its axis in an oscillatory motion, the outgoing section of pipe is retarded (or decelerated) and the return section is advanced (or accelerated), producing a twist in the pipe.

Coriolis Meter

The force is applied to oscillate the flow pipes and the Coriolis effect is the principle used to determine the acceleration due to the torque (the amount of twisting). Sensors are used to measure the amount of twist in the flowtubes within the meter as a result of the flowtube vibration and deflection due to the mass flow. The amount of twist measured is proportional to the mass flow rate and is measured by magnetic sensors mounted on the tubes.

Developments on the looped pipe Coriolis meter were made to keep to the pipes straight. This is done by making the pipes straight and parallel. The force is applied by oscillating the pipes at the resonant frequency. This has the advantage of reducing pressure loss in the pipeline.

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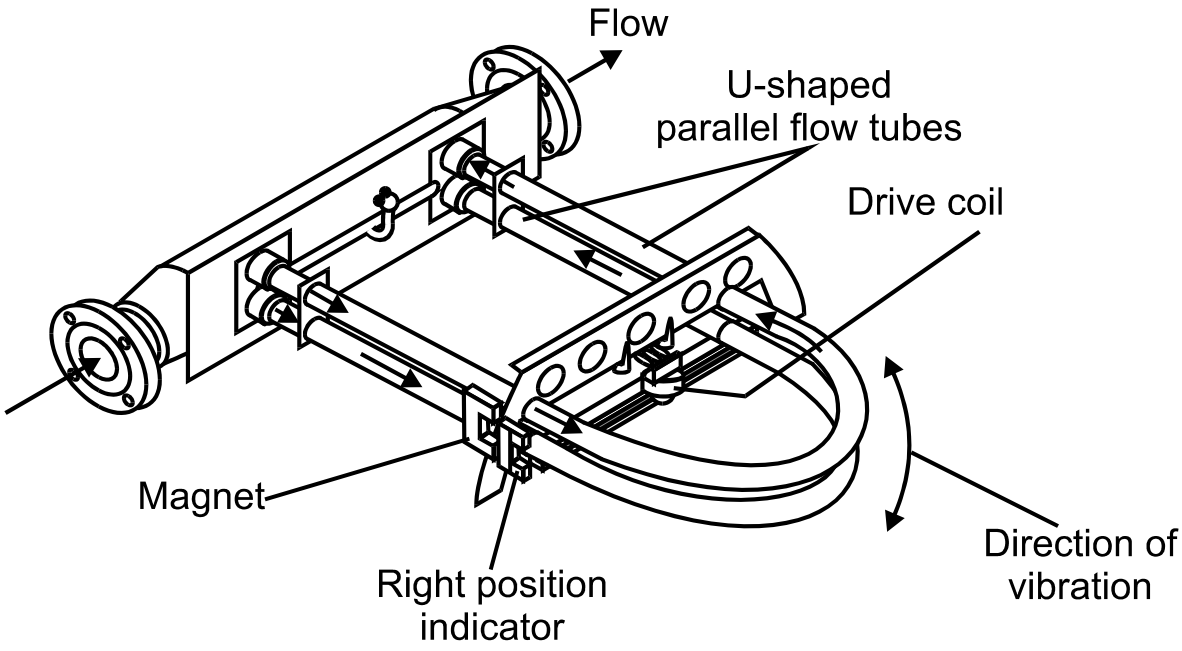


Figure 5.40
Coriolis meter construction

Straight Through Meter

A development to the looped typed Coriolis meter is the straight through pipe version, which has the added advantage of lower pressure loss.

The rotational movement in this type of meter is provided by vibrating the pipes and the Coriolis force develops in the pipes. The pipes are vibrated at their resonant frequency and sensors are used to detect the movement of the pipe. For no flow the sensors detect the same movement, however when liquid flows there is a difference between the oscillations of the two pipes. This is caused as the flow is accelerated on the inlet and decelerated on the outlet. As before with twist, this difference in the phase of the oscillations is proportional to mass flow.

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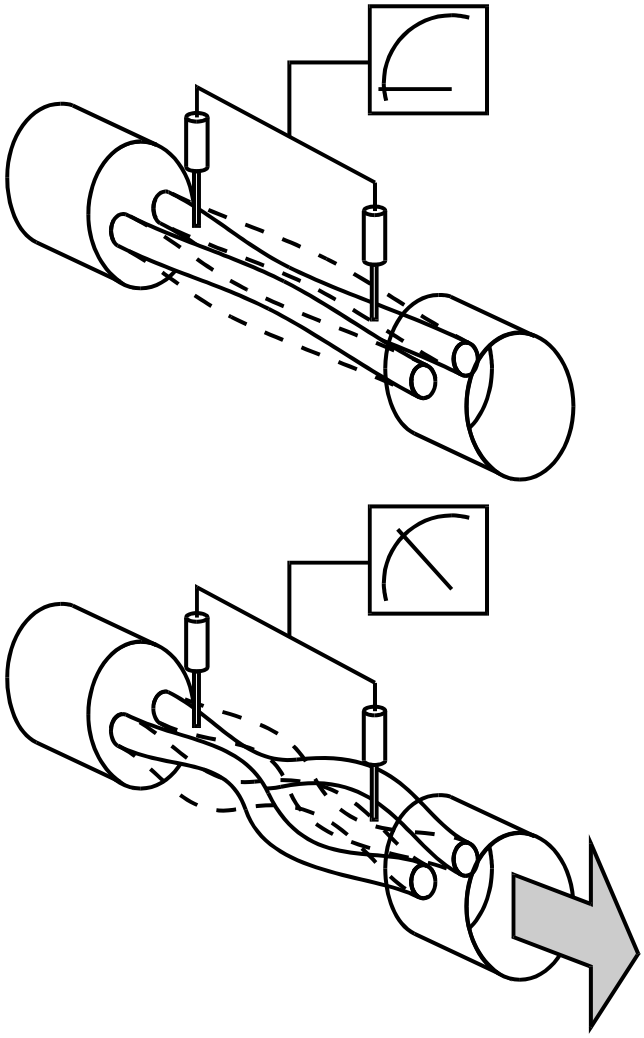


Figure 5.41
Basic principle of 'Straight Through' pipe (a) without fluid, and (b) with fluid flow
(Courtesy of Endress & Hauser)

Advantages

- Direct, in-line mass flow measurement.
- Independent of temperature, pressure, density, conductivity and viscosity.
- Sensor capable of transmitting mass flow, density and temperature information.
- High density capability.
- Conductivity independent.
- Suitable for hydrocarbon measurements.
- Suitable for density measurement.

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Disadvantages

- Cost.
- Affected by vibration.
- Installation costs.
- Adjustment of zero point.

Application Limitations

- High temperature.
- Vibration.
- Amount of gas in fluid.
- Restricted to low flow rates.
- Limited to pipe sizes of up to 150mm.

Summary

Coriolis meters provide direct, in-line and accurate mass flow measurements that are independent of temperature, pressure, viscosity and density. Mass flow, density and temperature can be accessed from the one sensor. They can also be used for almost any application when calibrated.

For critical control, mass flow rate is the preferred method of measurement and because of their accuracy Coriolis meters are becoming very common for applications requiring very tight control. Apart from custody transfer applications, they are used for chemical processes and expensive fluid handling.

5.9.2 Thermal Mass Flowmeters

The two main types of thermal mass flow measuring devices are:

- Thermal Anemometer
- Temperature rise flowmeter

Thermal anemometer

The thermal anemometer works by measuring the heat dissipation from a probe inserted in the line. The amount of heat taken from the probe is dependent on the fluid velocity and density, but is also a direct measure of the mass flow rate. The temperature is also measured for the calculation. They are also referred to as 'Hot wire probes'.

The probe can either be constant current or constant temperature.

In the constant current type, a fixed current is passed through the probe which causes heating in the probe. As the flow rate varies, so does the amount of heat taken from

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the probe and hence the temperature changes. The temperature is measured to derive the flow.

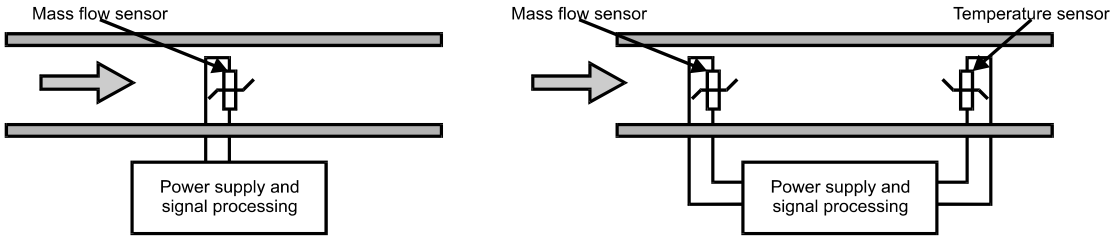


Figure 5.42
Thermal mass flowmeters

For the constant temperature type, a feedback loop is required to maintain a constant temperature. As the change of flow affects the temperature, the current needs to be regulated to maintain probe temperature. The flow rate is determined by the power required to heat the probe.

In comparing the constant current and constant temperature types it is shown that constant temperature devices have a faster response to flow changes.

These devices are primarily used for gases and are dependent on flow profile. They are therefore limited if the flow profile changes as they only measure at one point in the flowstream. Their limitations are similar to pitot tubes, which are also an insertion device into a flowstream.

To achieve laminar flow, 10 diameters of straight pipe are required upstream of the sensor. Later developments of these sensors incorporate a conditioning nozzle which concentrates the flow onto the sensor.

The temperature probe must protrude into the flowstream, and therefore may be easily damaged by corrosion and erosion. In addition the robustness of the system is compromised by the protrusions into the fluid stream, increasing the chances of leakage.

Advantages

- Fast response times, < 0.5milliseconds.

Disadvantages

- Require 10 diameters of straight pipe upstream.
- Have similar limitations to pitot tubes.

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Temperature rise flowmeter - insertion type

Temperature rise flowmeters work on the principle of heating the flowstream. By heating the flowstream at one point, the temperature can be measured both upstream and downstream of the heating point. Calculating the difference between the temperatures gives information about the flowrate.

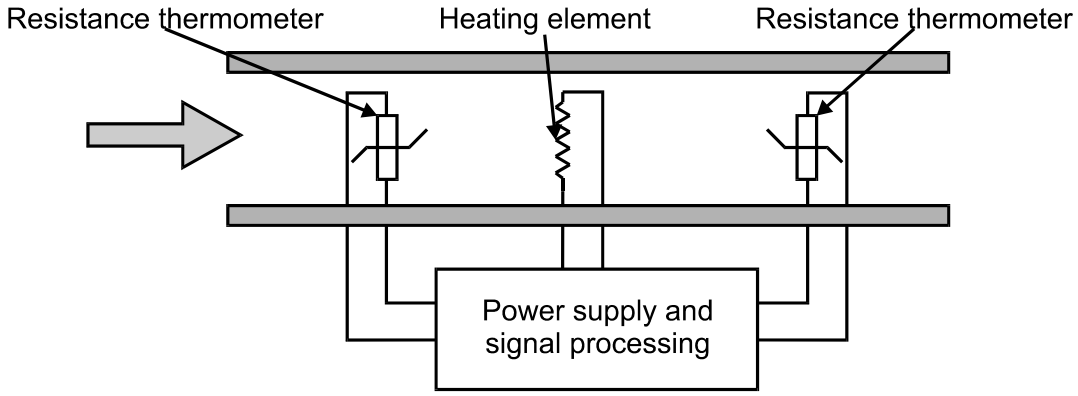


Figure 5.43
Principle of 'Temperature Rise' method

This method requires the measurement of actually heating the process fluid. It is therefore limited to gas applications at low flow rates.

As with the hot wire probe, the temperature sensors and the heater must protrude into the flowstream, and therefore may be easily damaged by corrosion and erosion. Also the robustness of the system is compromised by the protrusions into the fluid stream, increasing the chances of leakage.

Disadvantages

- Suitable for low gas flows only.
- Subject to erosion and corrosion.
- More tapping points, increased chances of leakage.

Temperature rise flowmeter - external type

Developments of the insertion type of sensing have moved the heating and sensing elements to outside the pipe to overcome the problems with tapping points. By limiting the tapping points, the chance of leakage (and associated maintenance) is greatly reduced, if not eliminated.

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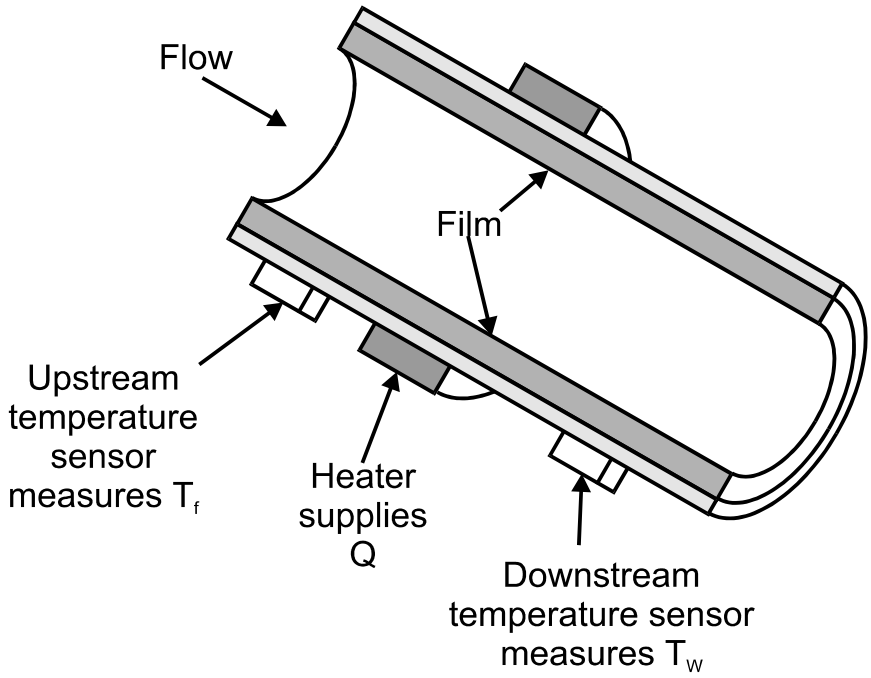


Figure 5.44
Thermal flowmeter with external elements and heater

This type of sensing mainly applies to small pipe diameters.

For larger pipe diameters, a sample of the flow can be taken and sensing applied in this way.

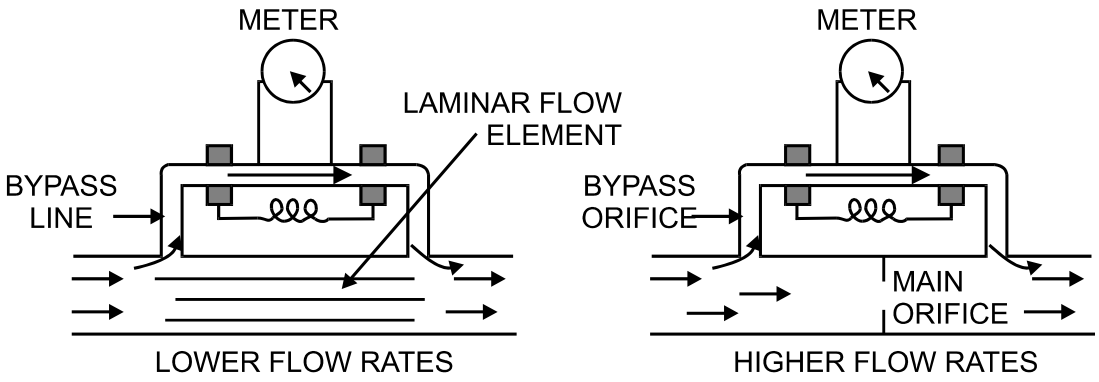


Figure 5.45
Bypass-type thermal mass flowmeter

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Advantages

- Non contact, non intrusive sensing
- No obstruction to flow
- Reduced maintenance

Disadvantages

- Suitable for low gas flows only
- Subject to erosion and corrosion

Summary

Thermal mass flowmeters are mainly used to measure the flow of clean gases with known heat capacities. They are commonly used in the refining and chemical industries.

5.9.3 Radiation Density

One of the limitations of flow meters is that they measure volumetric flow only. As mentioned previously, if the density is known and is constant, then mass flow rate can be calculated. Problems arise when the density varies.

It is however possible and quite acceptable practice to combine volumetric flow equipment with density measuring devices to obtain accurate mass flow measurements. One such combination is the use of a magmeter with a radiation densitometer.

5.10 Installation Considerations

Full Pipe Meters

It is often a requirement in flow measurement that the level in the pipe be maintained. Options are sometimes available for taking into account the level in the pipe, however this can prove to be an expensive alternative.

It is possible, at the expense of a very small added pressure loss, to lower the section of pipe containing the flowmeter so that the liquid pools in that area.

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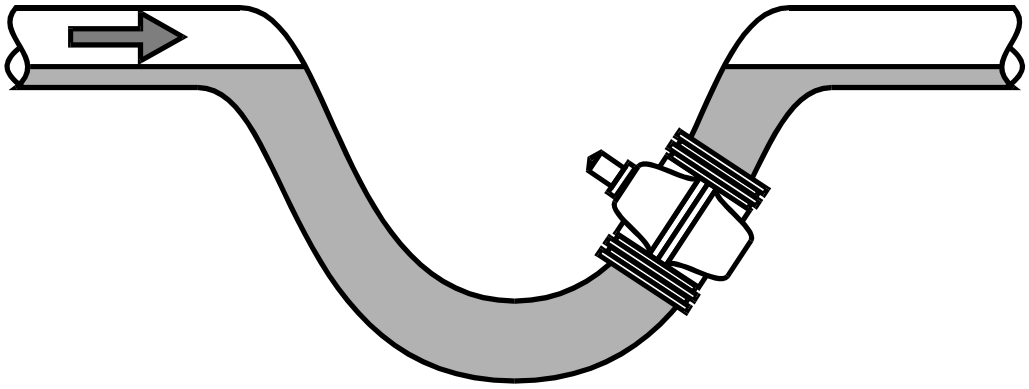


Figure 5.46
Flowmeter remains full when installed in (a) an invert, or (b) a V-section

5.11 Impact on the Overall Control Loop

There are two main problems at low flow rates:

- The minimum flow
- The increased error

One of the main problems with controlling a system from the flow rate is the range of control. The turndown defines the minimum flow compared with the maximum rated flow. Problems arise when controlling at or near the minimum flow, or if flow is required below the minimum flow rate.

Another problem is the increased error that results from low flow rates, particularly with differential pressure devices, where the flow rate is the square root of pressure.

5.11.1 The Minimum Flow

A single flow meter needs to be sized for the maximum flow of the application, however this may not allow suitable control at the lower flow rates. One method to increase the rangeability of the system is to use parallel piping. If the flow decreases then one of the runs can be closed to sustain an increased flow in the operating runs. This reduces the Rangeability required and maintains the highest accuracy of the system.

The accuracy of the measurement is optimised by maintaining the flowrate through the meters. The valves shown are for isolation of the separate runs.

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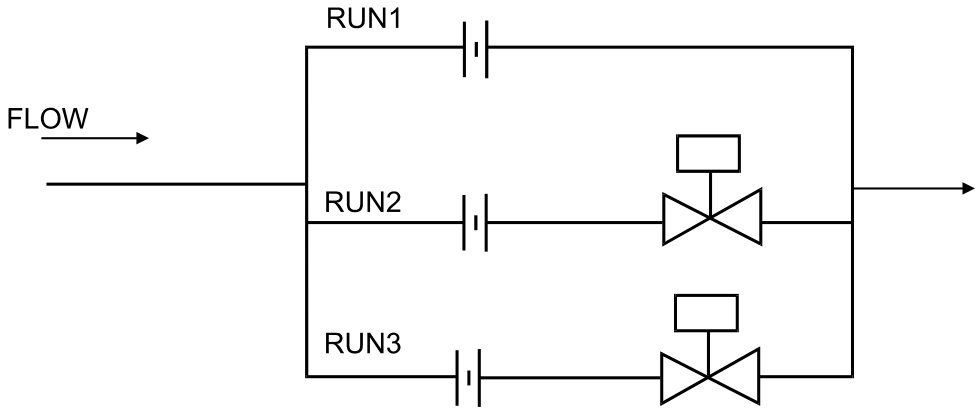


Figure 5.47
Parallel arrangement

The Increased Error

In relation to orifice meters, the orifice plate has an error of about 2% of the true flowrate. Problems arise with the differential pressure measurement as the error is in terms of percent of span.

Flow Rate	Max DP error
10	0.5%
7.5	1.0%
5	2.0%
3	5.0%

Table 5.9
Maximum differential pressure error

For higher turndown ratios to be achieved or maintained, the accuracy needs to be kept in check. The components of the accuracy are:

- Reference accuracy
- Ambient temperature effect
- Static pressure effect
- Time drift

The time drift can be reduced by regular calibration and the temperature effect can also be reduced using suitable enclosures or operating environments.

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5.12 Selection Tables

Flowmeter Technology	Accuracy (±)	Range-ability	Fluids	Pipe sizes (in.)	Max press (PSIG)	Temperature range (^oF)	Rel. Press. loss
Inferential Meters							
Orifice & DP xmtr	1-5%F	4:1	L,G,S	≥2	6000	≤1000	H
Integral orifice assembly	1-2%F	4:1	L,G,S	0.5-1.5	3000	-40-300	H
Variable area-purge	5-10%F	10:1	L,G	0.125-1.5	200	0-250	M
Variable area-glass tube	1-2%F	10:1	L,G	0.25-2	500	0-250	M
Variable area-metal tube	2-10%F	10:1	L,G	0.5-6	6000	≤1000	M
Velocity Meters							
Electromagnetic	0.25%-1%R	30:1	L	0.15-60	5000	≤350	L
Vortex	0.5%-1.25%R	20:1	L,G,S	1-8	1500	-150-800	M
Turbine	0.15-0.5%R	10:1	L	0.5-30	6000	-450-	M
Propeller	2%R	15:1	L	2-12	230	6000	M
Vane	2.5%R	10:1	L	1	5000	0-300	M
						≤180	
Mass Meters							
Coriolis	0.15%R	80:1	L,G,S	0.25-3	1450	-400-400	L
	0.2%R	20:1	L,G,S	0.06-6	5700	-400-400	M
	0.2%R	20:1	L,G,S	0.5-1.5	900	32-800	M
	0.2%R	20:1	L,G,S	0.25-2	1500	-400-400	M
Thermal	1%F	50:1	G	0.125-8	4500	32-150	L
	0.5%F	50:1	L	0.06-0.25	4500	40-165	L
Volumetric Meters							
BiRotor PD	0.15-0.5%R	10:1	L	1.5-16	1440	-20-450	M
Oval Gear PD	0.25-0.5%R	10:1	L	0.25-4	300	-40-600	M
Nutating disc	2.0%R	10:1	L	1.25	2000	32-200	H
Oscillating piston	0.5%R	5:1	L	0.5-2	400	≤300	H

Table 5.10
Flowmeter Selection Criteria
(Courtesy of Fisher Rosemount)

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5.13 Future Technologies

The newer technologies in flowmeters such as the Coriolis mass, ultrasonic and vortex are replacing the traditional differential pressure flowmeters and their orifice plates and other restrictive primary elements.

The selection of newer technologies, such as the coriolis mass flowmeter is primarily driven by the need for high accuracy in measurements. Ultrasonic flowmeters are becoming more popular due to their increased accuracy over previous versions. In addition they are non-intrusive and have a wide rangeability. Vortex flowmeters are also commonly used, as they are less expensive and can measure steam, liquid and gas.

5.13.1 Limiting Factors

The growth of the new technology is determined by a number of factors, such as:

1. Traditional technologies.

   Traditional technologies such as the orifice plate, turbine, and positive displacement meters are widely used in existing applications. Many users prefer to stay with a proven technology.

2. No reason to change

   Users generally require a compelling reason to change to a different or newer technology. As a result they usually choose the same type of flowmeter when selecting a replacement meter.

3. Familiarity

   It is quite often that the newer technologies are not well understood by the user.

4. Approvals

   Traditional technologies have a strong advantage in having approvals granted by industry associations.

5.13.2 Multivariable Flowmeters

Technological developments with pressure transmitters have seen the introduction of multivariable flowmeters that provide a measurement of the inferred mass flow. There is also a demand for improving the primary elements used with pressure transmitters.

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As previously mentioned, most volumetric flowmeters, such as differential pressure types, sense one process variable. This is used to calculate the volumetric flow or calculate the mass flow.

Multivariable flowmeters measure more than one process variable and can produce more than one output. A multivariable flowmeter may have sensors for differential pressure, static pressure and process temperature, which it uses to calculate mass flow.

Multivariable technology is not limited to differential pressure devices. Magnetic flowmeters can be used with process temperature measurement and a density compensation formula to calculate mass flow.

Devices such as these claim to measure mass flow, but the measurement obtained is a calculated or inferred value rather than a direct measurement, such as that obtained with a coriolis meter. Calculated mass flow is less accurate than a direct measurement because of the inaccuracies in the variables involved.

It is not uncommon for the primary elements to be purchased from one supplier, with the pressure transmitters purchased elsewhere. Manufacturers are recognising these limitations and are developing integral devices. By integrating a sensing element with a transmitter in a single unit, this system resembles most other flowmeters. This combination also simplifies installation and calibration procedures.

This approach sees more pressure sensing devices remaining in operation, with the ability of changing primary elements, or to change to multivariable flowmeters.

The limitation in sensing remains in the primary device. As flowmeters become more accurate, the electronics become quite precise in their interpretation and relaying of the signal, but are subject to the accuracy of the primary element.

The future of flow measurement is subject to the search and development of improved primary elements, whether they be existing or new technology.

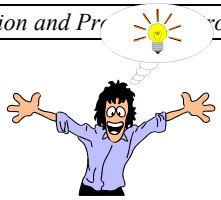
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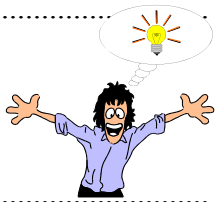


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